

Understanding NMR Spectroscopy

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2 NMR and energy levels

The picture that we use to understand most kinds of spectroscopy is that molecules have a set of *energy levels* and that the lines we see in spectra are due to *transitions* between these energy levels. Such a transition can be caused by a photon of light whose frequency, ν , is related to the energy gap, ΔE , between the two levels according to:

$$\Delta E = h\nu$$

where h is a universal constant known as Planck's constant. For the case shown in Fig. 2.1, $\Delta E = E_2 - E_1$.

In NMR spectroscopy we tend not to use this approach of thinking about energy levels and the transitions between them. Rather, we use different rules for working out the appearance of multiplets and so on. However, it is useful, especially for understanding more complex experiments, to think about how the familiar NMR spectra we see are related to energy levels. To start with we will look at the energy levels of just one spin and then move on quickly to look at two and three coupled spins. In such spin systems, as they are known, we will see that in principle there are other transitions, called *multiple quantum transitions*, which can take place. Such transitions are not observed in simple NMR spectra, but we can detect them indirectly using two-dimensional experiments; there are, as we shall see, important applications of such multiple quantum transitions.

Finally, we will look at *strongly coupled spectra*. These are spectra in which the simple rules used to construct multiplets no longer apply because the shift differences between the spins have become small compared to the couplings. The most familiar effect of strong coupling is the “roofing” or “tilting” of multiplets. We will see how such spectra can be analysed in some simple cases.

2.1 Frequency and energy: sorting out the units

NMR spectroscopists tend to use some rather unusual units, and so we need to know about these and how to convert from one to another if we are not to get into a muddle.

Chemical shifts

It is found to a very good approximation that the frequencies at which NMR absorptions (lines) occur scale linearly with the magnetic field strength. So, if the line from TMS comes out on one spectrometer at 400 MHz, doubling the magnetic field will result in it coming out at 800 MHz. If we wanted to quote the NMR frequency it would be inconvenient to have to specify the exact magnetic field strength as well. In addition, the numbers we would have

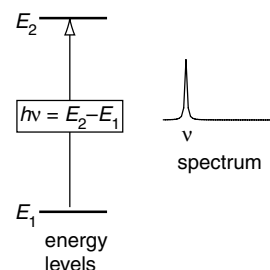


Fig. 2.1 A line in the spectrum is associated with a transition between two energy levels.

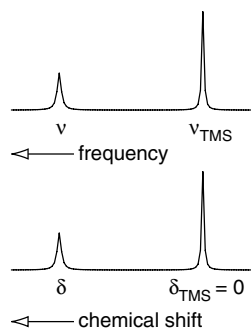


Fig. 2.2 An NMR spectrum can be plotted as a function of frequency, but it is more convenient to use the chemical shift scale in which frequencies are expressed relative to that of an agreed reference compound, such as TMS in the case of proton spectra.

to quote would not be very memorable. For example, would you like to quote the shift of the protons in benzene as 400.001234 MHz?

We neatly side-step both of these problems by quoting the *chemical shift* relative to an agreed *reference compound*. For example, in the case of proton NMR the reference compound is TMS. If the frequency of the line we are interested in is ν (in Hz) and the frequency of the line from TMS is ν_{TMS} (also in Hz), the chemical shift of the line is computed as:

$$\delta = \frac{\nu - \nu_{\text{TMS}}}{\nu_{\text{TMS}}}. \quad (2.1)$$

As all the frequencies scale with the magnetic field, this ratio is independent of the magnetic field strength. Typically, the chemical shift is rather small so it is common to multiply the value for δ by 10^6 and then quote its value in *parts per million*, or ppm. With this definition the chemical shift of the reference compound is 0 ppm.

$$\delta_{\text{ppm}} = 10^6 \times \frac{\nu - \nu_{\text{TMS}}}{\nu_{\text{TMS}}}. \quad (2.2)$$

Sometimes we want to convert from shifts in ppm to frequencies. Suppose that there are two peaks in the spectrum at shifts δ_1 and δ_2 in ppm. What is the frequency separation between the two peaks? It is easy enough to work out what it is in ppm, it is just $(\delta_2 - \delta_1)$. Writing this difference out in terms of the definition of chemical shift given in Eq. 2.2 we have:

$$\begin{aligned} (\delta_2 - \delta_1) &= 10^6 \times \frac{\nu_2 - \nu_{\text{TMS}}}{\nu_{\text{TMS}}} - 10^6 \times \frac{\nu_1 - \nu_{\text{TMS}}}{\nu_{\text{TMS}}} \\ &= 10^6 \times \frac{\nu_2 - \nu_1}{\nu_{\text{TMS}}}. \end{aligned}$$

Multiplying both sides by ν_{TMS} now gives us what we want:

$$(\nu_2 - \nu_1) = 10^{-6} \times \nu_{\text{TMS}} \times (\delta_2 - \delta_1).$$

It is often sufficiently accurate to replace ν_{TMS} with the spectrometer *reference frequency*, about which we will explain later.

If we want to change the ppm scale of a spectrum into a frequency scale we need to decide where zero is going to be. One choice for the zero frequency point is the line from the reference compound. However, there are plenty of other possibilities so it is as well to regard the zero point on the frequency scale as arbitrary.

Angular frequency

Frequencies are most commonly quoted in Hz, which is the same as “per second” or s^{-1} . Think about a point on the edge of a disc which is rotating about its centre. If the disc is moving at a constant speed, the point returns to the same position at regular intervals each time it has completed 360° of rotation. The time taken for the point to return to its original position is called the *period*, τ .

The frequency, ν , is simply the inverse of the period:

$$\nu = \frac{1}{\tau}.$$

For example, if the period is 0.001 s, the frequency is $1/0.001 = 1000$ Hz.

There is another way of expressing the frequency, which is in *angular units*. Recall that 360° is 2π radians. So, if the point completes a rotation in τ seconds, we can say that it has rotated through 2π radians in τ seconds. The angular frequency, ω , is given by

$$\omega = \frac{2\pi}{\tau}.$$

The units of this frequency are “radians per second” or rad s^{-1} . ν and ω are related via

$$\nu = \frac{\omega}{2\pi} \quad \text{or} \quad \omega = 2\pi\nu.$$

We will find that angular frequencies are often the most natural units to use in NMR calculations. Angular frequencies will be denoted by the symbols ω or Ω whereas frequencies in Hz will be denoted ν .

Energies

A photon of frequency ν has energy E given by

$$E = h\nu$$

where h is Planck’s constant. In SI units the frequency is in Hz and h is in J s^{-1} . If we want to express the frequency in angular units then the relationship with the energy is

$$\begin{aligned} E &= h \frac{\omega}{2\pi} \\ &= \hbar\omega \end{aligned}$$

where $\hbar$ (pronounced “h bar” or “h cross”) is Planck’s constant divided by 2π .

The point to notice here is that frequency, in either Hz or rad s^{-1} , is directly proportional to energy. So, there is really nothing wrong with quoting energies in frequency units. All we have to remember is that there is a factor of h or $\hbar$ needed to convert to Joules if we need to. It turns out to be much more convenient to work in frequency units throughout, and so this is what we will do. So, do not be concerned to see an energy expressed in Hz or rad s^{-1} .

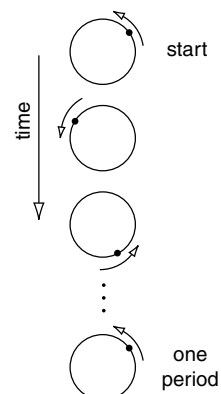


Fig. 2.3 A point at the edge of a circle which is moving at a constant speed returns to its original position after a time called the *period*. During each period the point moves through 2π radians or 360° .

2.2 Nuclear spin and spin states

NMR spectroscopy arises from the fact that nuclei have a property known as *spin*; we will not concern ourselves with where this comes from, but just take it as a fact. Quantum mechanics tells us that this nuclear spin is characterised by a nuclear spin quantum number, I . For all the nuclei that we are going

Strictly, α is the low energy state for nuclei with a positive gyromagnetic ratio, more of which below.

to be concerned with, $I = \frac{1}{2}$, although other values are possible. A spin-half nucleus has an interaction with a magnetic field which gives rise to two energy levels; these are characterised by another quantum number m which quantum mechanics tells us is restricted to the values $-I$ to I in integer steps. So, in the case of a spin-half, there are only two values of m , $-\frac{1}{2}$ and $+\frac{1}{2}$.

By tradition in NMR the energy level (or *state*, as it is sometimes called) with $m = \frac{1}{2}$ is denoted α and is sometimes described as “spin up”. The state with $m = -\frac{1}{2}$ is denoted β and is sometimes described as “spin down”. For the nuclei we are interested in, the α state is the one with the lowest energy.

If we have two spins in our molecule, then each spin can be in the α or β state, and so there are four possibilities: $\alpha_1\alpha_2$, $\alpha_1\beta_2$, $\beta_1\alpha_2$ and $\beta_1\beta_2$. These four possibilities correspond to four energy levels. Note that we have added a subscript 1 or 2 to differentiate the two nuclei, although often we will dispense with these and simply take it that the first spin state is for spin 1 and the second for spin 2. So $\alpha\beta$ implies $\alpha_1\beta_2$ etc.

We can continue the same process for three or more spins, and as each spin is added the number of possible combinations increases. So, for example, for three spins there are 8 combinations leading to 8 energy levels. It should be noted here that there is only a one-to-one correspondence between these spin state combinations and the energy levels in the case of weak coupling, which we will assume from now on. Further details are to be found in section 2.6.

2.3 One spin

There are just two energy levels for a single spin, and these can be labelled with either the m value of the labels α and β . From quantum mechanics we can show that the energies of these two levels, E_α and E_β , are:

$$E_\alpha = +\frac{1}{2}\nu_{0,1} \quad \text{and} \quad E_\beta = -\frac{1}{2}\nu_{0,1}$$

where $\nu_{0,1}$ is the *Larmor frequency* of spin 1 (we will need the 1 later on, but it is a bit superfluous here as we only have one spin). In fact it is easy to see that the energies just depend on m :

$$E_m = m\nu_{0,1}.$$

You will note here that, as explained in section 2.1 we have written the energies in frequency units. The Larmor frequency depends on a quantity known as the *gyromagnetic ratio*, γ , the chemical shift δ , and the strength of the applied magnetic field, B_0 :

$$\nu_{0,1} = -\frac{1}{2\pi}\gamma_1(1 + \delta_1)B_0 \quad (2.3)$$

where again we have used the subscript 1 to distinguish the of nucleus. The magnetic field is normally given in units of Tesla (symbol T). The gyromagnetic ratio is characteristic of a particular kind of nucleus, such as proton or carbon-13; its units are normally $\text{rad s}^{-1} \text{ T}^{-1}$. In fact, γ can be positive or negative; for the commonest nuclei (protons and carbon-13) it is positive. For such nuclei we therefore see that the Larmor frequency is negative.

This negative Larmor frequency for nuclei with a positive γ sometimes seems a bit unnatural and awkward, but to be consistent we need to stick with this convention. We will see in a later chapter that all this negative frequency really means is that the spin precesses in a particular sense.

To take a specific example, for protons $\gamma = +2.67 \times 10^8 \text{ rad s}^{-1} \text{ T}^{-1}$, so in a magnetic field of 4.7 T the Larmor frequency of a spin with chemical shift zero is

$$\begin{aligned}\nu_0 &= -\frac{1}{2\pi} \gamma (1 + \delta) B_0 \\ &= -\frac{1}{2\pi} \times 2.67 \times 10^8 \times 4.7 = -200 \times 10^6 \text{ Hz}.\end{aligned}$$

In other words, the Larmor frequency is -200 MHz .

We can also calculate the Larmor frequency in angular units, ω_0 , in which case the factor of $1/2\pi$ is not needed:

$$\omega_0 = -\gamma(1 + \delta)B_0$$

which gives a value of $1.255 \times 10^9 \text{ rad s}^{-1}$.

Spectrum

As you may know from other kinds of spectroscopy you have met, only certain transitions are *allowed* i.e. only certain ones actually take place. There are usually rules – called *selection rules* – about which transitions can take place; these rules normally relate to the quantum numbers which are characteristic of each state or energy level.

In the case of NMR, the selection rule refers to the quantum number m : only transitions in which m changes by one (up or down) are allowed. This is sometimes expressed as

$$\begin{aligned}\Delta m &= m(\text{initial state}) - m(\text{final state}) \\ &= \pm 1.\end{aligned}$$

Another way as saying this is that one spin can flip between “up” and “down” or vice versa.

In the case of a single spin-half, the change in m between the two states is $(+\frac{1}{2} - (-\frac{1}{2})) = 1$ so the transition is allowed. We can now simply work out the frequency of the allowed transition:

$$\begin{aligned}\nu_{\alpha\beta} &= E_\beta - E_\alpha \\ &= -\frac{1}{2}\nu_{0,1} - (+\frac{1}{2}\nu_{0,1}) \\ &= -\nu_{0,1}.\end{aligned}$$

Note that we have taken the energy of the upper state minus that of the lower state. In words, therefore, we see one transition at the minus the Larmor frequency, $-\nu_{0,1}$.

You would be forgiven for thinking that this is all an enormous amount of effort to come up with something very simple! However, the techniques and ideas developed in this section will enable us to make faster progress with the case of two and three coupled spins, which we consider next.

2.4 Two spins

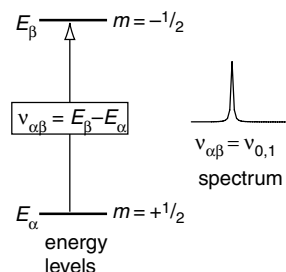


Fig. 2.4 The transition between the two energy levels of a spin-half is allowed, and results in a single line at the Larmor frequency of the spin.

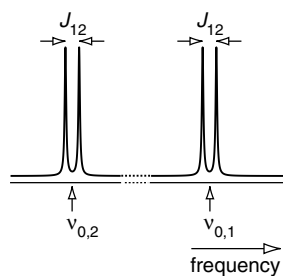


Fig. 2.5 Schematic spectrum of two coupled spins showing two doublets with equal splittings. As indicated by the dashed lines, the separation of the Larmor frequencies is much larger than the coupling between the spins.

We know that the spectrum of two coupled spins consists of two doublets, each split by the same amount, one centred at the chemical shift of the first spin and one at the shift of the second. The splitting of the doublets is the scalar coupling, J_{12} , quoted in Hz; the subscripts indicate which spins are involved. We will write the shifts of the two spins as δ_1 and δ_2 , and give the corresponding Larmor frequencies, $\nu_{0,1}$ and $\nu_{0,2}$ as:

$$\nu_{0,1} = -\frac{1}{2\pi}\gamma_1(1 + \delta_1)B_0$$

$$\nu_{0,2} = -\frac{1}{2\pi}\gamma_2(1 + \delta_2)B_0.$$

If the two nuclei are of the same type, such a proton, then the two gyromagnetic ratios are equal; such a two spin system would be described as *homonuclear*. The opposite case is where the two nuclei are of different types, such as proton and carbon-13; such a spin system is described as *heteronuclear*.

Energy levels

As was already described in section 2.2, there are four possible combinations of the spin states of two spins and these combinations correspond to four energy levels. Their energies are given in the following table:

number	spin states	energy
1	$\alpha\alpha$	$+\frac{1}{2}\nu_{0,1} + \frac{1}{2}\nu_{0,2} + \frac{1}{4}J_{12}$
2	$\alpha\beta$	$+\frac{1}{2}\nu_{0,1} - \frac{1}{2}\nu_{0,2} - \frac{1}{4}J_{12}$
3	$\beta\alpha$	$-\frac{1}{2}\nu_{0,1} + \frac{1}{2}\nu_{0,2} - \frac{1}{4}J_{12}$
4	$\beta\beta$	$-\frac{1}{2}\nu_{0,1} - \frac{1}{2}\nu_{0,2} + \frac{1}{4}J_{12}$

The second column gives the spin states of spins 1 and 2, in that order. It is easy to see that these energies have the general form:

$$E_{m_1 m_2} = m_1 \nu_{0,1} + m_2 \nu_{0,2} + m_1 m_2 J_{12}$$

where m_1 and m_2 are the m values for spins 1 and 2, respectively.

For a homonuclear system $\nu_{0,1} \approx \nu_{0,2}$; also both Larmor frequencies are much greater in magnitude than the coupling (the Larmor frequencies are of the order of hundreds of MHz, while couplings are at most a few tens of Hz). Therefore, under these circumstances, the energies of the $\alpha\beta$ and $\beta\alpha$ states are rather similar, but very different from the other two states. For a heteronuclear system, in which the Larmor frequencies differ significantly, the four levels are all at markedly different energies. These points are illustrated in Fig. 2.6.

Spectrum

The selection rule is the same as before, but this time it applies to the quantum number M which is found by adding up the m values for each of the spins. In this case:

$$M = m_1 + m_2.$$

The resulting M values for the four levels are:

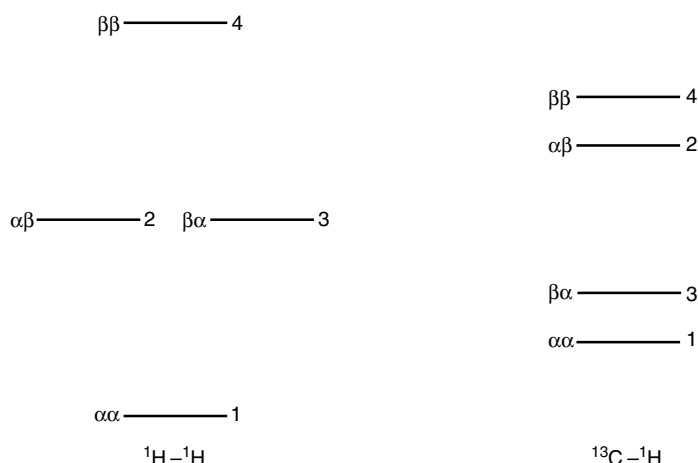


Fig. 2.6 Energy levels, drawn approximately to scale, for two spin systems. On the left is shown a homonuclear system (two protons); on this scale the $\alpha\beta$ and $\beta\alpha$ states have the same energy. On the right is the case for a carbon-13 – proton pair. The Larmor frequency of proton is about four times that of carbon-13, and this is clear reflected in the diagram. The $\alpha\beta$ and $\beta\alpha$ states now have substantially different energies.

number	spin states	M
1	$\alpha\alpha$	1
2	$\alpha\beta$	0
3	$\beta\alpha$	0
4	$\beta\beta$	-1

The selection rule is that $\Delta M = \pm 1$, i.e. the value of M can change up or down by one unit. This means that the allowed transitions are between levels 1 & 2, 3 & 4, 1 & 3 and 2 & 4. The resulting frequencies are easily worked out; for example, the 1–2 transition:

$$\begin{aligned}
 \nu_{12} &= E_2 - E_1 \\
 &= +\frac{1}{2}\nu_{0,1} - \frac{1}{2}\nu_{0,2} - \frac{1}{4}J_{12} - (\frac{1}{2}\nu_{0,1} + \frac{1}{2}\nu_{0,2} + \frac{1}{4}J_{12}) \\
 &= -\nu_{0,2} - \frac{1}{2}J_{12}.
 \end{aligned}$$

Throughout we will use the convention that when computing the transition frequency we will take the energy of the upper state minus the energy of the lower: $\Delta E = E_{\text{upper}} - E_{\text{lower}}$.

The complete set of transitions are:

transition	spin states	frequency
1 $\rightarrow$ 2	$\alpha\alpha \rightarrow \alpha\beta$	$-\nu_{0,2} - \frac{1}{2}J_{12}$
3 $\rightarrow$ 4	$\beta\alpha \rightarrow \beta\beta$	$-\nu_{0,2} + \frac{1}{2}J_{12}$
1 $\rightarrow$ 3	$\alpha\alpha \rightarrow \beta\alpha$	$-\nu_{0,1} - \frac{1}{2}J_{12}$
2 $\rightarrow$ 4	$\alpha\beta \rightarrow \beta\beta$	$-\nu_{0,1} + \frac{1}{2}J_{12}$

The energy levels and corresponding schematic spectrum are shown in Fig. 2.7. There is a lot we can say about this spectrum. Firstly, each allowed transition corresponds to one of the spins flipping from one spin state to the other, while the spin state of the other spin remains fixed. For example, transition 1–2 involves a spin 2 going from α to β whilst spin 1 remains in the α state. In this transition we say that spin 2 is *active* and spin 1 is *passive*. As

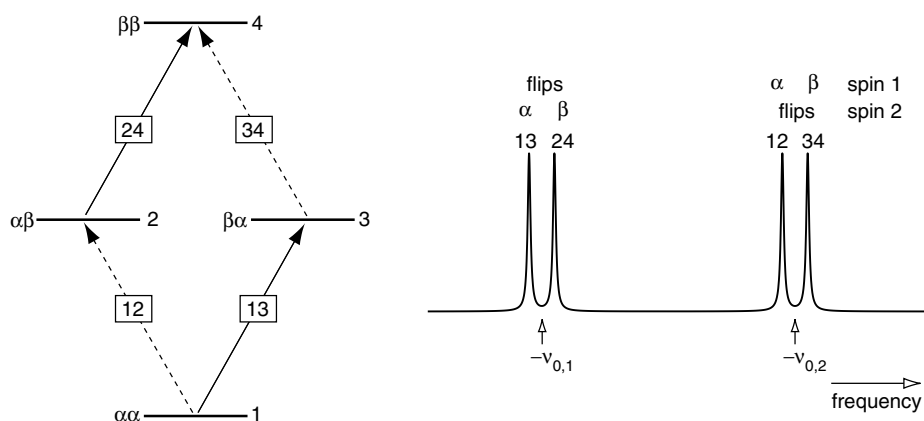


Fig. 2.7 On the left, the energy levels of a two-spin system; the arrows show the allowed transitions: solid lines for transitions in which spin 1 flips and dotted for those in which spin 2 flips. On the right, the corresponding spectrum; it is assumed that the Larmor frequency of spin 2 is greater in magnitude than that of spin 1 and that the coupling J_{12} is positive.

spin 2 flips in this transition, it is not surprising that the transition forms one part of the doublet for spin 2.

Transition 3–4 is similar to 1–2 except that the passive spin (spin 1) is in the β state; this transition forms the second line of the doublet for spin 2. This discussion illustrates a very important point, which is that the lines of a multiplet can be associated with different spin states of the coupled (passive) spins. We will use this kind of interpretation very often, especially when considering two-dimensional spectra.

The two transitions in which spin 1 flips are 1–3 and 2–4, and these are associated with spin 2 being in the α and β spin states, respectively. Which spin flips and the spins states of the passive spins are shown in Fig. 2.7.

What happens if the coupling is negative? If you work through the table you will see that there are still four lines at the same frequencies as before. All that changes is the labels of the lines. So, for example, transition 1–2 is now the right line of the doublet, rather than the left line. From the point of view of the spectrum, what swaps over is the spin state of the passive spin associated with each line of the multiplet. The overall appearance of the spectrum is therefore independent of the sign of the coupling constant.

Multiple quantum transitions

There are two more transitions in our two-spin system which are not allowed by the usual selection rule. The first is between states 1 and 4 ($\alpha\alpha \rightarrow \beta\beta$) in which both spins flip. The ΔM value is 2, so this is called a *double-quantum* transition. Using the same terminology, all of the allowed transitions described above, which have $\Delta M = 1$, are *single-quantum* transitions. From the table of energy levels it is easy to work out that its frequency is $(-\nu_{0,1} - \nu_{0,2})$ i.e. the sum of the Larmor frequencies. Note that the coupling has no effect on the frequency of this line.

The second transition is between states 2 and 3 ($\alpha\beta \rightarrow \beta\alpha$); again, both

spins flip. The ΔM value is 0, so this is called a *zero-quantum* transition, and its frequency is $(-\nu_{0,1} + \nu_{0,2})$ i.e. the difference of the Larmor frequencies. As with the double-quantum transition, the coupling has no effect on the frequency of this line.

In a two spin system the double- and zero-quantum spectra are not especially interesting, but we will see in a three-spin system that the situation is rather different. We will also see later on that in two-dimensional spectra we can exploit to our advantage the special properties of these multiple-quantum spectra.

2.5 Three spins

If we have three spins, each of which is coupled to the other two, then the spectrum consists of three doublets of doublets, one centred at the shift of each of the three spins; the *spin topology* is shown in Fig. 2.9. The appearance of these multiplets will depend on the relative sizes of the couplings. For example, if $J_{12} = J_{13}$ the doublet of doublets from spin 1 will collapse to a 1:2:1 triplet. On the other hand, if $J_{12} = 0$, only doublets will be seen for spin 1 and spin 2, but spin 3 will still show a doublet of doublets.

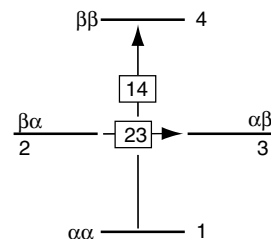


Fig. 2.8 In a two-spin system there is one double quantum transition (1–4) and one zero-quantum transition (2–3); the frequency of neither of these transitions are affected by the size of the coupling between the two spins.

2.7 Exercises

E 2-1

In a proton spectrum the peak from TMS is found to be at 400.135705 MHz. What is the shift, in ppm, of a peak which has a frequency of 400.136305 MHz? Recalculate the shift using the spectrometer frequency, ν_{spec} quoted by the manufacturer as 400.13 MHz rather than ν_{TMS} in the denominator of Eq. 2.2:

$$\delta_{\text{ppm}} = 10^6 \times \frac{\nu - \nu_{\text{TMS}}}{\nu_{\text{spec}}}.$$

Does this make a significant difference to the value of the shift?

E 2-2

Two peaks in a proton spectrum are found at 1.54 and 5.34 ppm. The spectrometer frequency is quoted as 400.13 MHz. What is the separation of these two lines in Hz and in rad s^{-1} ?

E 2-3

Calculate the Larmor frequency (in Hz and in rad s^{-1}) of a carbon-13 resonance with chemical shift 48 ppm when recorded in a spectrometer with a magnetic field strength of 9.4 T. The gyromagnetic ratio of carbon-13 is $+6.7283 \times 10^7 \text{ rad s}^{-1} \text{ T}^{-1}$.

E 2-4

Of course in reality the Larmor frequencies are to be tens or hundreds of MHz, not 100 Hz! However, it makes the numbers easier to handle if we use these unrealistic small values; the principles remain the same, however.

Consider a system of two weakly coupled spins. Let the Larmor frequency of the first spin be -100 Hz and that of the second spin be -200 Hz , and let the coupling between the two spins be -5 Hz . Compute the frequencies of the lines in the normal (single quantum) spectrum.

Make a sketch of the spectrum, roughly to scale, and label each line with the energy levels involved (i.e. 1-2 etc.). Also indicate for each line which spin flips and the spin state of the passive spin. Compare your sketch with Fig. 2.7 and comment on any differences.

E 2-5

For a three spin system, draw up a table similar to that on page 2-10 showing the frequencies of the four lines of the multiplet from spin 2. Then, taking $\nu_{0,2} = -200 \text{ Hz}$, $J_{23} = 4 \text{ Hz}$ and the rest of the parameters as in Fig. 2.11, compute the frequencies of the lines which comprise the spin 2 multiplet. Make a sketch of the multiplet (roughly to scale) and label the lines in the same way as is done in Fig. 2.11. How would these labels change if $J_{23} = -4 \text{ Hz}$?

On an energy level diagram, indicate the four transitions which comprise the spin 2 multiplet, and which four comprise the spin 3 multiplet.

E 2-6

For a three spin system, compute the frequencies of the six zero-quantum

transitions and also mark these on an energy level diagram. Do these six transitions fall into natural groups? How would you describe the spectrum?

E 2-7

Calculate the line frequencies and intensities of the spectrum for a system of two spins with the following parameters: $\nu_{0,1} = -10$ Hz, $\nu_{0,2} = -20$ Hz, $J_{12} = 5$ Hz. Make a sketch of the spectrum (roughly to scale) indicating which transition is which and the position of the Larmor frequencies.

E 2-8

The spectrum from a strongly-coupled two spin system showed lines at the following frequencies, in Hz, (intensities are given in brackets): 32.0 (1.3), 39.0 (0.7), 6.0 (0.7), 13.0 (1.3). Determine the values of the coupling constant and the two Larmor frequencies. Show that the values you find are consistent with the observed intensities.

Make sure that you have your calculator set to "radians" when you compute $\sin 2\theta$.

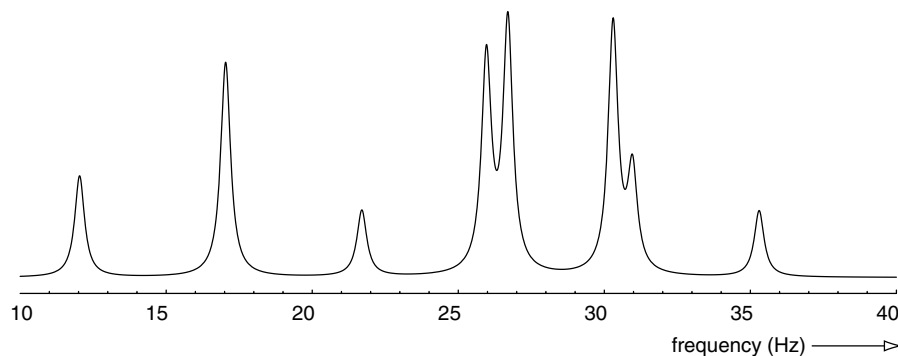


Fig. 2.18 The AB part of an ABX spectrum

E 2-9

Figure 2.18 shows the AB part of an ABX spectrum. Disentangle the two subspectra, mark in the rough positions of the effective Larmor frequencies and hence estimate the size of the AX and BX couplings. Also, give the value of the AB coupling.

3 The vector model

For most kinds of spectroscopy it is sufficient to think about energy levels and selection rules; this is not true for NMR. For example, using this energy level approach we cannot even describe how the most basic pulsed NMR experiment works, let alone the large number of subtle two-dimensional experiments which have been developed. To make any progress in understanding NMR experiments we need some more tools, and the first of these we are going to explore is the *vector model*.

This model has been around as long as NMR itself, and not surprisingly the language and ideas which flow from the model have become the language of NMR to a large extent. In fact, in the strictest sense, the vector model can only be applied to a surprisingly small number of situations. However, the ideas that flow from even this rather restricted area in which the model can be applied are carried over into more sophisticated treatments. It is therefore essential to have a good grasp of the vector model and how to apply it.

3.1 Bulk magnetization

We commented before that the nuclear spin has an interaction with an applied magnetic field, and that it is this which gives rise to the energy levels and ultimately an NMR spectrum. In many ways, it is permissible to think of the nucleus as behaving like a small bar magnet or, to be more precise, a *magnetic moment*. We will not go into the details here, but note that the quantum mechanics tells us that the magnetic moment can be aligned in any direction¹.

In an NMR experiment, we do not observe just one nucleus but a very large number of them (say 10^{20}), so what we need to be concerned with is the net effect of all these nuclei, as this is what we will observe.

If the magnetic moments were all to point in random directions, then the small magnetic field that each generates will cancel one another out and there will be no net effect. However, it turns out that at equilibrium the magnetic moments are not aligned randomly but in such a way that when their contributions are all added up there is a net magnetic field along the direction of the applied field (B_0). This is called the *bulk magnetization* of the sample.

The magnetization can be represented by a vector – called the *magnetization vector* – pointing along the direction of the applied field (z), as shown in Fig. 3.1. From now on we will only be concerned with what happens to this vector.

This would be a good point to comment on the axis system we are going to use – it is called a *right-handed set*, and such a set of axes is shown in Fig. 3.1. The name right-handed comes about from the fact that if you imagine grasping

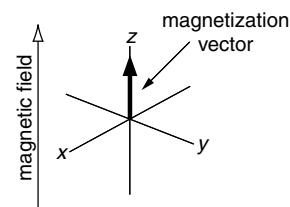


Fig. 3.1 At equilibrium, a sample has a net magnetization along the magnetic field direction (the z axis) which can be represented by a magnetization vector. The axis set in this diagram is a right-handed one, which is what we will use throughout these lectures.

¹It is a common misconception to state that the magnetic moment must either be aligned with or against the magnetic field. In fact, quantum mechanics says no such thing (see Levitt Chapter 9 for a very lucid discussion of this point).

the z axis with your right hand, your fingers curl from the x to the y axes.

3.2 Larmor precession

Suppose that we have managed, some how, to tip the magnetization vector away from the z axis, such that it makes an angle β to that axis. We will see later on that such a tilt can be brought about by a radiofrequency pulse. Once tilted away from the z axis we find is that the magnetization vector rotates about the direction of the magnetic field sweeping out a cone with a constant angle; see Fig. 3.2. The vector is said to *precesses* about the field and this particular motion is called *Larmor precession*.

If the magnetic field strength is B_0 , then the frequency of the Larmor precession is ω_0 (in rad s^{-1})

$$\omega_0 = -\gamma B_0$$

or if we want the frequency in Hz, it is given by

$$\nu_0 = -\frac{1}{2\pi}\gamma B_0$$

where γ is the gyromagnetic ratio. These are of course exactly the same frequencies that we encountered in section 2.3. In words, the frequency at which the magnetization precesses around the B_0 field is exactly the same as the frequency of the line we see from the spectrum on one spin; this is no accident.

As was discussed in section 2.3, the Larmor frequency is a signed quantity and is negative for nuclei with a positive gyromagnetic ratio. This means that for such spins the precession frequency is negative, which is precisely what is shown in Fig. 3.2.

We can sort out positive and negative frequencies in the following way. Imagine grasping the z axis with your right hand, with the thumb pointing along the $+z$ direction. The fingers then curl in the sense of a positive precession. Inspection of Fig. 3.2 will show that the magnetization vector is rotating in the opposite sense to your fingers, and this corresponds to a negative Larmor frequency.

3.3 Detection

The precession of the magnetization vector is what we actually detect in an NMR experiment. All we have to do is to mount a small coil of wire round the sample, with the axis of the coil aligned in the xy -plane; this is illustrated in Fig. 3.3. As the magnetization vector “cuts” the coil a current is induced which we can amplify and then record – this is the so-called *free induction* signal which is detected in a pulse NMR experiment. The whole process is analogous to the way in which electric current can be generated by a magnet rotating inside a coil.

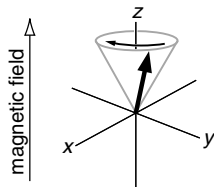


Fig. 3.2 If the magnetization vector is tilted away from the z axis it executes a precessional motion in which the vector sweeps out a cone of constant angle to the magnetic field direction. The direction of precession shown is for a nucleus with a positive gyromagnetic ratio and hence a negative Larmor frequency.

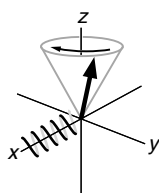


Fig. 3.3 The precessing magnetization will cut a coil wound round the x axis, thereby inducing a current in the coil. This current can be amplified and detected; it is this that forms the free induction signal. For clarity, the coil has only been shown on one side of the x axis.

Essentially, the coil detects the x -component of the magnetization. We can easily work out what this will be. Suppose that the equilibrium magnetization vector is of size M_0 ; if this has been tilted through an angle β towards the x axis, the x -component is $M_0 \sin \beta$; Fig. 3.4 illustrates the geometry.

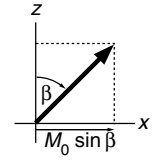


Fig. 3.4 Tilting the magnetization through an angle θ gives an x -component of size $M_0 \sin \beta$.

Although the magnetization vector precesses on a cone, we can visualize what happens to the x - and y -components much more simply by just thinking about the projection onto the xy -plane. This is shown in Fig. 3.5.

At time zero, we will assume that there is only an x -component. After a time τ_1 the vector has rotated through a certain angle, which we will call ϵ_1 . As the vector is rotating at ω_0 radians per second, in time τ_1 the vector has moved through $(\omega_0 \times \tau_1)$ radians; so $\epsilon_1 = \omega_0 \tau_1$. At a later time, say τ_2 , the vector has had longer to precess and the angle ϵ_2 will be $(\omega_0 \tau_2)$. In general, we can see that after time t the angle is $\epsilon = \omega_0 t$.

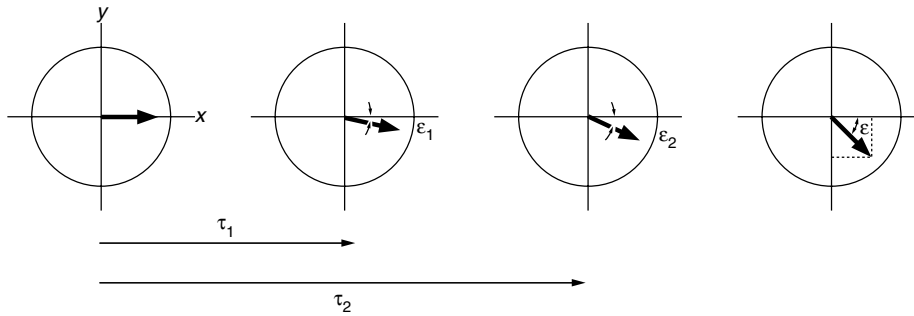


Fig. 3.5 Illustration of the precession of the magnetization vector in the xy -plane. The angle through which the vector has precessed is given by $\omega_0 t$. On the right-hand diagram we see the geometry for working out the x and y components of the vector.

We can now easily work out the x - and y -components of the magnetization using simple geometry; this is illustrated in Fig. 3.5. The x -component is proportional to $\cos \epsilon$ and the y -component is negative (along $-y$) and proportional to $\sin \epsilon$. Recalling that the initial size of the vector is $M_0 \sin \beta$, we can deduce that the x - and y -components, M_x and M_y respectively, are:

$$M_x = M_0 \sin \beta \cos(\omega_0 t)$$

$$M_y = -M_0 \sin \beta \sin(\omega_0 t).$$

Plots of these signals are shown in Fig. 3.6. We see that they are both simple oscillations at the Larmor frequency. Fourier transformation of these signals gives us the familiar spectrum – in this case a single line at ω_0 ; the details of how this works will be covered in a later chapter. We will also see in a later section that in practice we can easily detect *both* the x - and y -components of the magnetization.

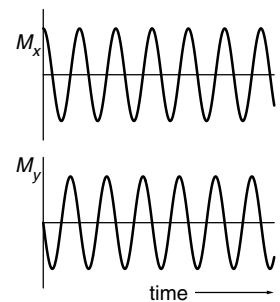


Fig. 3.6 Plots of the x - and y -components of the magnetization predicted using the approach of Fig. 3.5. Fourier transformation of these signals will give rise to the usual spectrum.

3.4 Pulses

We now turn to the important question as to how we can rotate the magnetization away from its equilibrium position along the z axis. Conceptually it is

easy to see what we have to do. All that is required is to (suddenly) replace the magnetic field along the z axis with one in the xy -plane (say along the x axis). The magnetization would then precess about the new magnetic field which would bring the vector down away from the z axis, as illustrated in Fig. 3.7.

Unfortunately it is all but impossible to switch the magnetic field suddenly in this way. Remember that the main magnetic field is supplied by a powerful superconducting magnet, and there is no way that this can be switched off; we will need to find another approach, and it turns out that the key is to use the idea of *resonance*.

The idea is to apply a very small magnetic field along the x axis but one which is oscillating at or near to the Larmor frequency – that is *resonant* with the Larmor frequency. We will show that this small magnetic field is able to rotate the magnetization away from the z axis, even in the presence of the very strong applied field, B_0 .

Conveniently, we can use the same coil to generate this oscillating magnetic field as the one we used to detect the magnetization (Fig. 3.3). All we do is feed some radiofrequency (RF) power to the coil and the resulting oscillating current creates an oscillating magnetic field along the x -direction. The resulting field is called the *radiofrequency* or *RF field*. To understand how this weak RF field can rotate the magnetization we need to introduce the idea of the *rotating frame*.

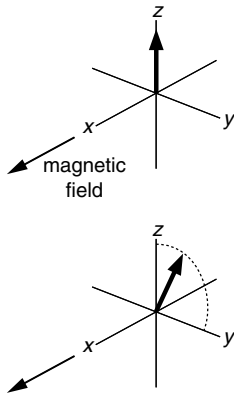


Fig. 3.7 If the magnetic field along the z axis is replaced quickly by one along x , the magnetization will then precess about the x axis and so move towards the transverse plane.

Rotating frame

When RF power is applied to the coil wound along the x axis the result is a magnetic field which oscillates along the x axis. The magnetic field moves back and forth from $+x$ to $-x$ passing through zero along the way. We will take the frequency of this oscillation to be ω_{RF} (in rad s^{-1}) and the size of the magnetic field to be $2B_1$ (in T); the reason for the 2 will become apparent later. This frequency is also called the *transmitter frequency* for the reason that a radiofrequency transmitter is used to produce the power.

It turns out to be a lot easier to work out what is going on if we replace, in our minds, this *linearly* oscillating field with two counter-rotating fields; Fig. 3.8 illustrates the idea. The two counter rotating fields have the same magnitude B_1 . One, denoted B_1^+ , rotates in the positive sense (from x to y) and the other, denoted B_1^- , rotates in the negative sense; both are rotating at the transmitter frequency ω_{RF} .

At time zero, they are both aligned along the x axis and so add up to give a total field of $2B_1$ along the x axis. As time proceeds, the vectors move away from x , in opposite directions. As the two vectors have the same magnitude and are rotating at the same frequency the y -components *always cancel* one another out. However, the x -components shrink towards zero as the angle through which the vectors have rotated approaches $\frac{1}{2}\pi$ radians or 90° . As the angle increases beyond this point the x -component grows once more, but this time along the $-x$ axis, reaching a maximum when the angle of rotation is π . The fields continue to rotate, causing the x -component to drop back to zero and rise again to a value $2B_1$ along the $+x$ axis. Thus we see that the two

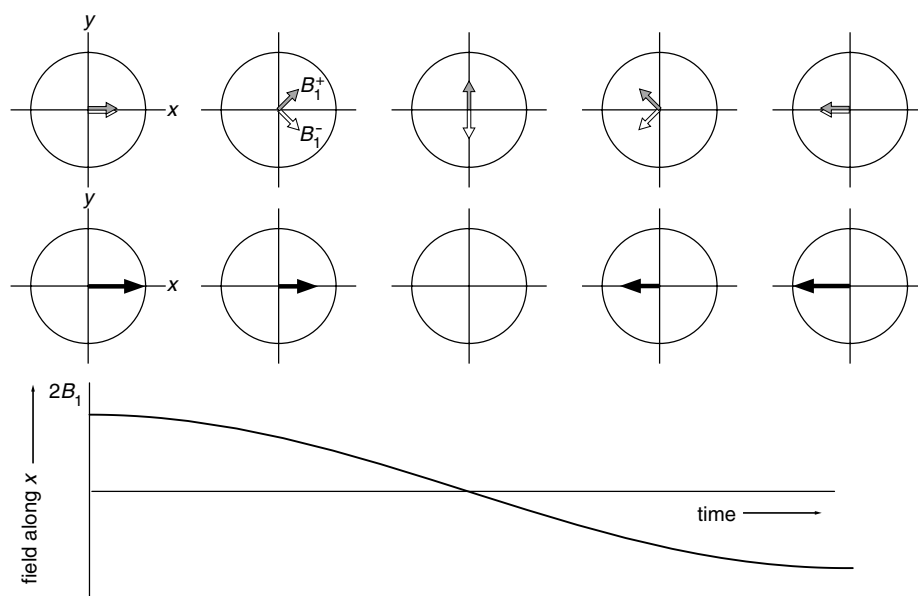


Fig. 3.8 Illustration of how two counter-rotating fields (shown in the upper part of the diagram and marked B_1^+ and B_1^-) add together to give a field which is oscillating along the x axis (shown in the lower part). The graph at the bottom shows how the field along x varies with time.

counter-rotating fields add up to the linearly oscillating one.

Suppose now that we think about a nucleus with a positive gyromagnetic ratio; recall that this means the Larmor frequency is negative so that the sense of precession is from x towards $-y$. This is the same direction as the rotation of B_1^- . It turns out that the other field, which is rotating in the opposite sense to the Larmor precession, has no significant interaction with the magnetization and so from now on we will ignore it.

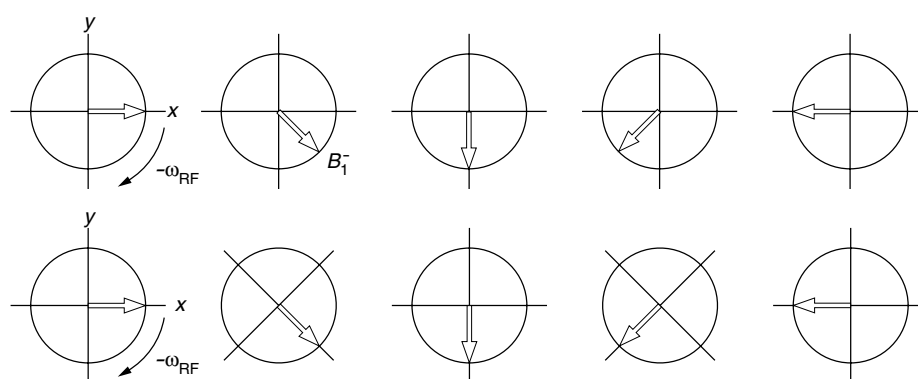


Fig. 3.9 The top row shows a field rotating at $-\omega_{\text{RF}}$ when viewed in a fixed axis system. The same field viewed in a set of axes rotating at $-\omega_{\text{RF}}$ appears to be static.

We now employ a mathematical trick which is to move to a co-ordinate system which, rather than being static (called the *laboratory frame*) is rotating about the z axis in the same direction and at the same rate as B_1^- (i.e. at $-\omega_{\text{RF}}$). In this rotating set of axes, or *rotating frame*, B_1^- appears to be static

and directed along the x axis of the rotating frame, as is shown in Fig. 3.9. This is a very nice result as the time dependence has been removed from the problem.

Larmor precession in the rotating frame

We need to consider what happens to the Larmor precession of the magnetization when this is viewed in this rotating frame. In the fixed frame the precession is at ω_0 , but suppose that we choose the rotating frame to be at the same frequency. In the rotating frame the magnetization will appear *not to move* i.e. the apparent Larmor frequency will be zero! It is clear that moving to a rotating frame has an effect on the *apparent* Larmor frequency.

The general case is when the rotating frame is at frequency $\omega_{\text{rot. fram.}}$; in such a frame the Larmor precession will appear to be at $(\omega_0 - \omega_{\text{rot. fram.}})$. This difference frequency is called the offset and is given the symbol Ω :

$$\Omega = \omega_0 - \omega_{\text{rot. fram.}} \quad (3.1)$$

We have used several times the relationship between the magnetic field and the precession frequency:

$$\omega = -\gamma B. \quad (3.2)$$

From this it follows that if the apparent Larmor frequency in the rotating frame is different from that in fixed frame it must also be the case that the *apparent* magnetic field in the rotating frame must be different from the actual applied magnetic field. We can use Eq. 3.2 to compute the apparent magnetic field, given the symbol ΔB , from the apparent Larmor frequency, Ω :

$$\begin{aligned} \Omega &= -\gamma \Delta B \\ \text{hence } \Delta B &= -\frac{\Omega}{\gamma}. \end{aligned}$$

This apparent magnetic field in the rotating frame is usually called the *reduced field*, ΔB .

If we choose the rotating frame to be at the Larmor frequency, the offset Ω will be zero and so too will the reduced field. This is the key to how the very weak RF field can affect the magnetization in the presence of the much stronger B_0 field. In the rotating frame this field along the z axis appears to shrink, and under the right conditions can become small enough that the RF field is dominant.

The effective field

From the discussion so far we can see that when an RF field is being applied there are two magnetic fields in the rotating frame. First, there is the RF field (or B_1 field) of magnitude B_1 ; we will make this field static by choosing the rotating frame frequency to be equal to $-\omega_{\text{RF}}$. Second, there is the reduced field, ΔB , given by $(-\Omega/\gamma)$. Since $\Omega = (\omega_0 - \omega_{\text{rot. fram.}})$ and $\omega_{\text{rot. fram.}} = -\omega_{\text{RF}}$ it follows that the offset is

$$\begin{aligned} \Omega &= \omega_0 - (-\omega_{\text{RF}}) \\ &= \omega_0 + \omega_{\text{RF}}. \end{aligned}$$

In this discussion we will assume that the gyromagnetic ratio is positive so that the Larmor frequency is negative.

This looks rather strange, but recall that ω_0 is negative, so if the transmitter frequency and the Larmor frequency are comparable the offset will be small.

In the rotating frame, the reduced field (which is along z) and the RF or B_1 field (which is along x) add vectorially to give an effective field, B_{eff} as illustrated in Fig. 3.10. The size of this effective field is given by:

$$B_{\text{eff}} = \sqrt{B_1^2 + \Delta B^2}. \quad (3.3)$$

The magnetization precesses about this effective field at frequency ω_{eff} given by

$$\omega_{\text{eff}} = \gamma B_{\text{eff}}$$

just in the same way that the Larmor precession frequency is related to B_0 .

By making the offset small, or zero, the effective field lies close to the xy -plane, and so the magnetization will be rotated from z down to the plane, which is exactly what we want to achieve. The trick is that although B_0 is much larger than B_1 we can affect the magnetization with B_1 by making it oscillate close to the Larmor frequency. This is the phenomena of resonance.

The angle between ΔB and B_{eff} is called the *tilt angle* and is usually given the symbol θ . From Fig. 3.10 we can see that:

$$\sin \theta = \frac{B_1}{B_{\text{eff}}} \quad \cos \theta = \frac{\Delta B}{B_{\text{eff}}} \quad \tan \theta = \frac{B_1}{\Delta B}.$$

All three definitions are equivalent.

The effective field in frequency units

For practical purposes the thing that is important is the precession frequency about the effective field, ω_{eff} . It is therefore convenient to think about the construction of the effective field not in terms of magnetic fields but in terms of the precession frequencies that they cause.

For each field the precession frequency is proportional to the magnetic field; the constant of proportion is γ , the gyromagnetic ratio. For example, we have already seen that in the rotating frame the apparent Larmor precession frequency, Ω , depends on the reduced field:

$$\Omega = -\gamma \Delta B.$$

We define ω_1 as the precession frequency about the B_1 field (the positive sign is intentional):

$$\omega_1 = \gamma B_1$$

and we already have

$$\omega_{\text{eff}} = \gamma B_{\text{eff}}.$$

Using these definitions in Eq. 3.3 ω_{eff} can be written as

$$\omega_{\text{eff}} = \sqrt{\omega_1^2 + \Omega^2}.$$

Figure 3.10 can be redrawn in terms of frequencies, as shown in Fig. 3.11. Similarly, the tilt angle can be expressed in terms of these frequencies:

$$\sin \theta = \frac{\omega_1}{\omega_{\text{eff}}} \quad \cos \theta = \frac{\Omega}{\omega_{\text{eff}}} \quad \tan \theta = \frac{\omega_1}{\Omega}.$$

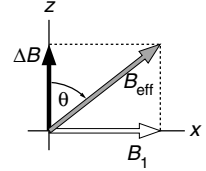


Fig. 3.10 In the rotating frame the effective field B_{eff} is the vector sum of the reduced field ΔB and the B_1 field. The tilt angle, θ , is defined as the angle between ΔB and B_{eff} .

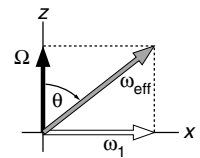


Fig. 3.11 The effective field can be thought of in terms of frequencies instead of the fields used in Fig 3.10.

On-resonance pulses

The simplest case to deal with is where the transmitter frequency is exactly the same as the Larmor frequency – it is said that the pulse is exactly *on resonance*. Under these circumstances the offset, Ω , is zero and so the reduced field, ΔB , is also zero. Referring to Fig. 3.10 we see that the effective field is therefore the same as the B_1 field and lies along the x axis. For completeness we also note that the tilt angle, θ , of the effective field is $\pi/2$ or 90° .

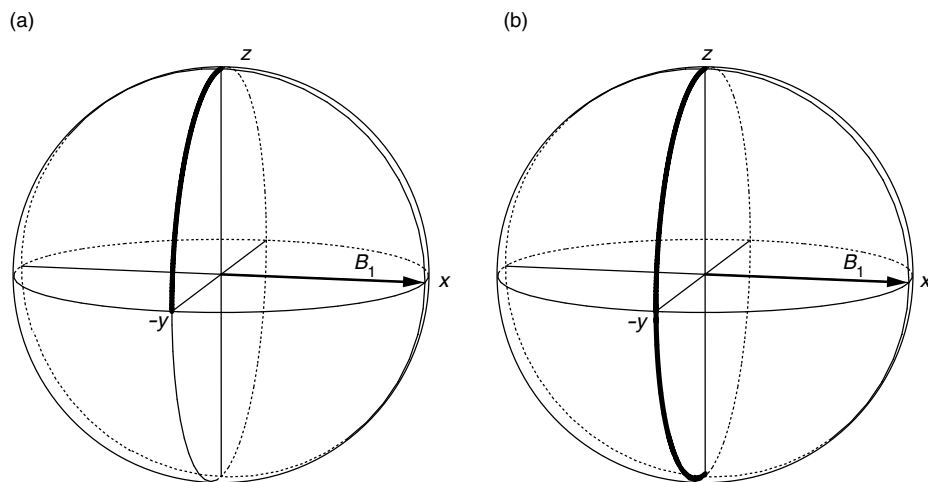


Fig. 3.12 A “grapefruit” diagram in which the thick line shows the motion of a magnetization vector during an on-resonance pulse. The magnetization is assumed to start out along $+z$. In (a) the pulse flip angle is 90° . The effective field lies along the x axis and so the magnetization precesses in the yz -plane. The rotation is in the positive sense about x so the magnetization moves toward the $-y$ axis. In (b) the pulse flip angle is 180° and so the magnetization ends up along $-z$.

In this situation the motion of the magnetization vector is very simple. Just as in Fig. 3.7 the magnetization precesses about the field, thereby rotating in the zy -plane. As we have seen above the precession frequency is ω_1 . If the RF field is applied for a time t_p , the angle, β , through which the magnetization has been rotated will be given by

$$\beta = \omega_1 t_p.$$

β is called the *flip angle of the pulse*. By altering the time for which the pulse has been applied we can alter the angle through which the magnetization is rotated.

In many experiments the commonly used flip angles are $\pi/2$ (90°) and π (180°). The motion of the magnetization vector during on-resonance 90° and 180° pulses are shown in Fig. 3.12. The 90° pulse rotates the magnetization from the equilibrium position to the $-y$ axis; this is because the rotation is in the positive sense. Imagine grasping the axis about which the rotation is taking place (the x axis) with your right hand; your fingers then curl in the sense of a positive rotation.

If the pulse flip angle is set to 180° the magnetization is taken all the way from $+z$ to $-z$; this is called an *inversion pulse*. In general, for a flip angle β

simple geometry tells us that the z - and y -components are

$$M_z = M_0 \cos \beta \quad M_y = -M_0 \sin \beta;$$

this is illustrated in Fig. 3.13.

Hard pulses

In practical NMR spectroscopy we usually have several resonances in the spectrum, each of which has a different Larmor frequency; we cannot therefore be on resonance with all of the lines in the spectrum. However, if we make the RF field strong enough we can effectively achieve this condition.

By “hard” enough we mean that the B_1 field has to be large enough that it is much greater than the size of the reduced field, ΔB . If this condition holds, the effective field lies along B_1 and so the situation is identical to the case of an on-resonance pulse. Thinking in terms of frequencies this condition translates to ω_1 being greater in magnitude than the offset, Ω .

It is often relatively easy to achieve this condition. For example, consider a proton spectrum covering about 10 ppm; if we put the transmitter frequency at about 5 ppm, the maximum offset is 5 ppm, either positive or negative; this is illustrated in Fig. 3.14. If the spectrometer frequency is 500 MHz the maximum offset is $5 \times 500 = 2500$ Hz. A typical spectrometer might have a 90° pulse lasting $12 \mu\text{s}$. From this we can work out the value of ω_1 . We start from

$$\beta = \omega_1 t_p \text{ hence } \omega_1 = \frac{\beta}{t_p}.$$

We know that for a 90° pulse $\beta = \pi/2$ and the duration, t_p is 12×10^{-6} s; therefore

$$\begin{aligned} \omega_1 &= \frac{\pi/2}{12 \times 10^{-6}} \\ &= 1.3 \times 10^5 \text{ rad s}^{-1}. \end{aligned}$$

The maximum offset is 2500 Hz, which is $2\pi \times 2500 = 1.6 \times 10^4 \text{ rad s}^{-1}$. We see that the RF field is about eight times the offset, and so the pulse can be regarded as strong over the whole width of the spectrum.

3.5 Detection in the rotating frame

To work out what is happening during an RF pulse we need to work in the rotating frame, and we have seen that to get this simplification the frequency of the rotating frame must match the transmitter frequency, ω_{RF} . Larmor precession can be viewed just as easily in the laboratory and rotating frames; in the rotating frame the precession is at the offset frequency, Ω .

It turns out that because of the way the spectrometer works the signal that we detect appears to be that in the rotating frame. So, rather than detecting an oscillation at the Larmor frequency, we see an oscillation at the offset, Ω .²

²Strictly this is only true if we set the receiver reference frequency to be equal to the transmitter frequency; this is almost always the case. More details will be given in Chapter 5.

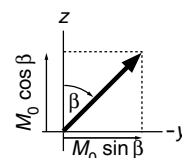


Fig. 3.13 If the pulse flip angle is β we can use simple geometry to work out the y - and z -components.

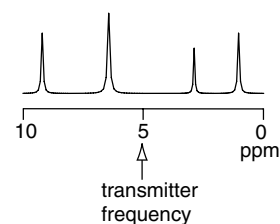


Fig. 3.14 Illustration of the range of offsets one might see in a typical proton spectrum. If the transmitter frequency is placed as shown, the maximum offset of a resonance will be 5 ppm.

It also turns out that we can detect both the x - and y -components of the magnetization in the rotating frame. From now on we will work exclusively in the rotating frame.

3.6 The basic pulse–acquire experiment

At last we are in a position to describe how the simplest NMR experiment works – the one we use every day to record spectra. The experiment comes in three periods:

1. The sample is allowed to come to equilibrium.
2. RF power is switched on for long enough to rotate the magnetization through 90° i.e. a 90° pulse is applied.
3. After the RF power is switched off we start to detect the signal which arises from the magnetization as it rotates in the transverse plane.

The timing diagram – or *pulse sequence* as it is usually known – is shown in Fig. 3.15.

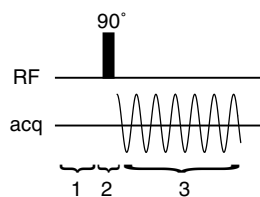


Fig. 3.15 Timing diagram or pulse sequence for the simple pulse–acquire experiment. The line marked “RF” shows the location of the pulses, and the line marked “acq” shows when the signal is recorded or acquired.

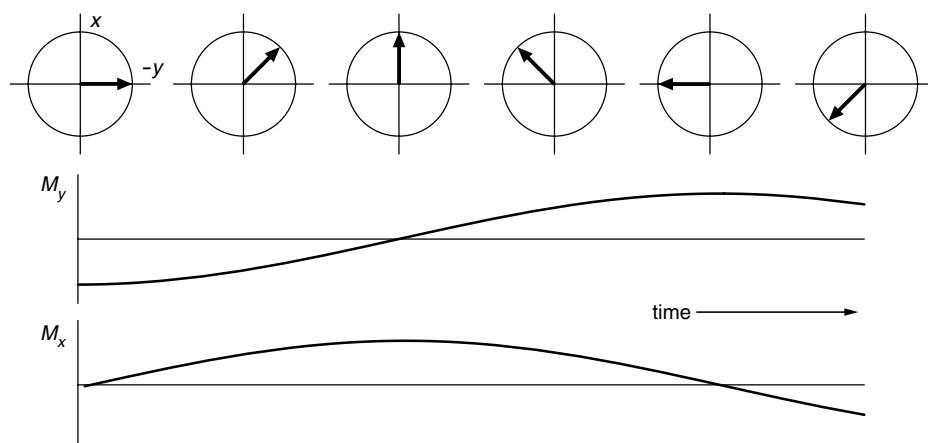


Fig. 3.16 Evolution during the acquisition time (period 3) of the pulse–acquire experiment. The magnetization starts out along $-y$ and evolves at the offset frequency, Ω (here assumed to be positive). The resulting x - and y -magnetizations are shown below.

During period 1 equilibrium magnetization builds up along the z axis. As was described above, the 90° pulse rotates this magnetization onto the $-y$ axis; this takes us to the end of period 2. During period 3 the magnetization precesses in the transverse plane at the offset Ω ; this is illustrated in Fig. 3.16. Some simple geometry, shown in Fig. 3.17, enables us to deduce how the x - and y -magnetizations vary with time. The offset is Ω so after time t the vector has precessed through an angle $(\Omega \times t)$. The y -component is therefore proportional to $\cos \Omega t$ and the x -component to $\sin \Omega t$. In full the signals are:

$$M_y = -M_0 \cos(\Omega t)$$

$$M_x = M_0 \sin(\Omega t).$$

As we commented on before, Fourier transformation of these signals will give the usual spectrum, with a peak appearing at frequency Ω .

Spectrum with several lines

If the spectrum has more than one line, then to a good approximation we can associate a magnetization vector with each. Usually we wish to observe all the lines at once, so we choose the B_1 field to be strong enough that for the range of offsets that these lines cover all the associated magnetization vectors will be rotated onto the $-y$ axis. During the acquisition time each precesses at its own offset, so the detected signal will be:

$$M_y = -M_{0,1} \cos \Omega_1 t - M_{0,2} \cos \Omega_2 t - M_{0,3} \cos \Omega_3 t \dots$$

where $M_{0,1}$ is the equilibrium magnetization of spin 1, Ω_1 is its offset and so on for the other spins. Fourier transformation of the free induction signal will produce a spectrum with lines at Ω_1 , Ω_2 etc.

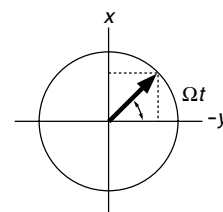


Fig. 3.17 The magnetization starts out along the $-y$ axis and rotates through an angle Ωt during time t .

3.7 Pulse calibration

It is crucial that the pulses we use in NMR experiments have the correct flip angles. For example, to obtain the maximum intensity in the pulse-acquire experiment we must use a 90° pulse, and if we wish to invert magnetization we must use a 180° pulse. Pulse calibration is therefore an important preliminary to any experiment.

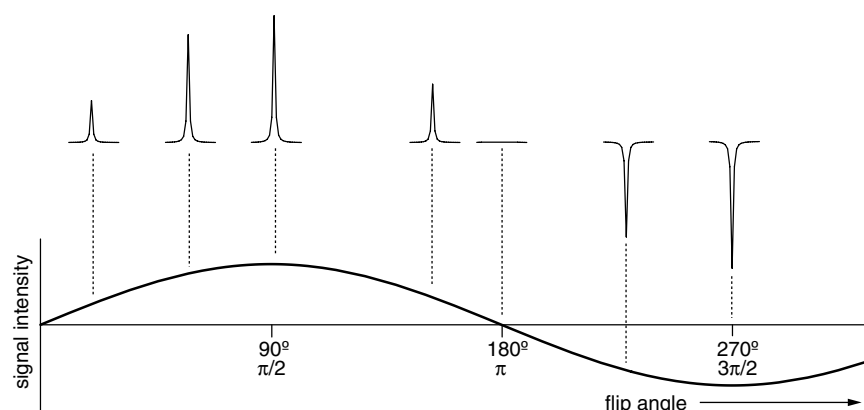


Fig. 3.18 Illustration of how pulse calibration is achieved. The signal intensity varies as $(\sin \beta)$ as shown by the curve at the bottom of the picture. Along the top are the spectra which would be expected for various different flip angles (indicated by the dashed lines). The signal is a maximum for a flip angle of 90° and goes through a null at 180° ; after that, the signal goes negative.

If we imagine an on-resonance or hard pulse we have already determined from Fig. 3.13 that the y -component of magnetization after a pulse of flip angle β is proportional to $\sin \beta$. If we therefore do a pulse-acquire experiment (section 3.6) and vary the flip angle of the pulse, we should see that the intensity of the signal varies as $\sin \beta$. A typical outcome of such an experiment is shown in Fig. 3.18.

The normal practice is to increase the flip angle until a null is found; the flip angle is then 180° . The reason for doing this is that the null is sharper than the maximum. Once the length of a 180° pulse is found, simply halving the time gives a 90° pulse.

Suppose that the 180° pulse was found to be of duration t_{180} . Since the flip angle is given by $\beta = \omega_1 t_p$ we can see that for a 180° pulse in which the flip angle is π

$$\pi = \omega_1 t_{180}$$

$$\text{hence } \omega_1 = \frac{\pi}{t_{180}}.$$

In this way we can determine ω_1 , usually called the RF *field strength* or the B_1 *field strength*.

It is usual to quote the field strength not in rad s^{-1} but in Hz, in which case we need to divide by 2π :

$$(\omega_1/2\pi) = \frac{1}{2t_{180}} \text{ Hz.}$$

For example, let us suppose that we found the null condition at $15.5 \mu\text{s}$; thus

$$\omega_1 = \frac{\pi}{t_{180}} = \frac{\pi}{15.5 \times 10^{-6}} = 2.03 \times 10^5 \text{ rad s}^{-1}.$$

In frequency units the calculation is

$$(\omega_1/2\pi) = \frac{1}{2t_{180}} = \frac{1}{2 \times 15.5 \times 10^{-6}} = 32.3 \text{ kHz.}$$

In normal NMR parlance we would say “the B_1 field is 32.3 kHz”. This is mixing the units up rather strangely, but the meaning is clear once you know what is going on!

3.8 The spin echo

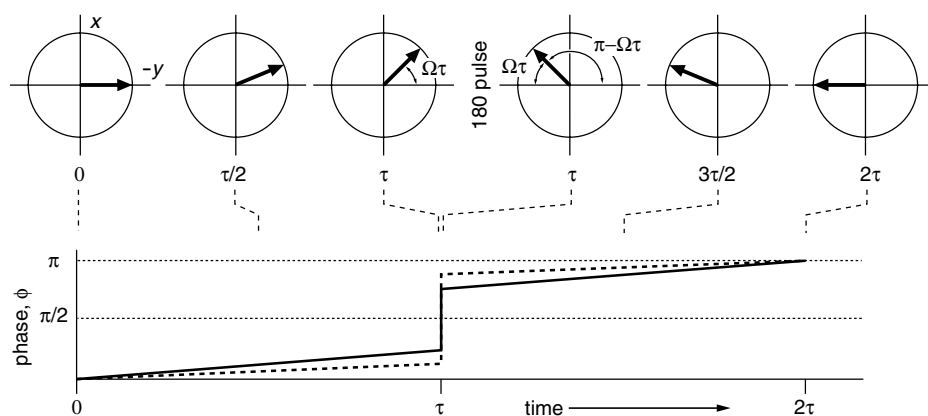


Fig. 3.19 Vector diagrams showing how a spin echo refocuses the evolution of the offset; see text for details. Also shown is a phase evolution diagram for two different offsets (the solid and the dashed line).

We are now able to analyse the most famous pulsed NMR experiment, the *spin echo*, which is a component of a very large number of more complex experiments. The pulse sequence is quite simple, and is shown in Fig. 3.20. The special thing about the spin echo sequence is that at the end of the second τ delay the magnetization ends up along the *same* axis, *regardless* of the values of τ and the offset, Ω .

We describe this outcome by saying that “the offset has been refocused”, meaning that at the end of the sequence it is just as if the offset had been zero and hence there had been no evolution of the magnetization. Figure 3.19 illustrates how the sequence works after the initial 90° pulse has placed the magnetization along the $-y$ axis.

During the first delay τ the vector precesses from $-y$ towards the x axis. The angle through which the vector rotates is simply $(\Omega\tau)$, which we can describe as a *phase*, ϕ . The effect of the 180° pulse is to move the vector to a mirror image position, with the mirror in question being in the xz -plane. So, the vector is now at an angle $(\Omega\tau)$ to the y axis rather than being at $(\Omega\tau)$ to the $-y$ axis.

During the second delay τ the vector continues to evolve; during this time it will rotate through a further angle of $(\Omega\tau)$ and therefore at the end of the second delay the vector will be aligned along the y axis. A few moments thought will reveal that as the angle through which the vector rotates during the first τ delay must be *equal* to that through which it rotates during the second τ delay; the vector will therefore always end up along the y axis *regardless of the offset*, Ω .

The 180° pulse is called a *refocusing pulse* because of the property that the evolution due to the offset during the first delay τ is refocused during the second delay. It is interesting to note that the spin echo sequence gives exactly the same result as the sequence $90^\circ - 180^\circ$ with the delays omitted.

Another way of thinking about the spin echo is to plot a phase evolution diagram; this is done at the bottom of Fig. 3.19. Here we plot the phase, ϕ , as a function of time. During the first τ delay the phase increases linearly with time. The effect of the 180° pulse is to change the phase from $(\Omega\tau)$ to $(\pi - \Omega\tau)$; this is the jump on the diagram at time τ . Further evolution for time τ causes the phase to increase by $(\Omega\tau)$ leading to a final phase at the end of the second τ delay of π . This conclusion is independent of the value of the offset Ω ; the diagram illustrates this by the dashed line which represents the evolution of vector with a smaller offset.

As has already been mentioned, the effect of the 180° pulse is to reflect the vectors in the xz -plane. The way this works is illustrated in Fig. 3.21. The arc through which the vectors are moved is different for each, but all the vectors end up in mirror image positions.

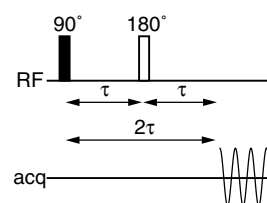


Fig. 3.20 Pulse sequence for the *spin echo* experiment. The 180° pulse (indicated by an open rectangle as opposed to the closed one for a 90° pulse) is in the centre of a delay of duration 2τ , thus separating the sequence into two equal periods, τ . The signal is acquired after the second delay τ , or put another way, when the time from the beginning of the sequence is 2τ . The durations of the pulses are in practice very much shorter than the delays τ but for clarity the length of the pulses has been exaggerated.

3.9 Pulses of different phases

So far we have assume that the B_1 field is applied along the x axis; this does not have to be so, and we can just as easily apply it along the y axis, for example. A pulse about y is said to be “phase shifted by 90° ” (we take an

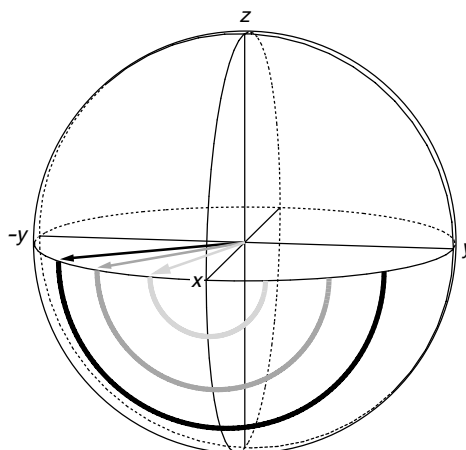


Fig. 3.21 Illustration of the effect of a 180° pulse on three vectors which start out at different angles from the $-y$ axis (coloured in black, grey and light grey). All three are rotated by 180° about the x axis on the trajectories indicated by the thick lines which dip into the southern hemisphere. As a result, the vectors end up in mirror image positions with respect to the xz -plane.

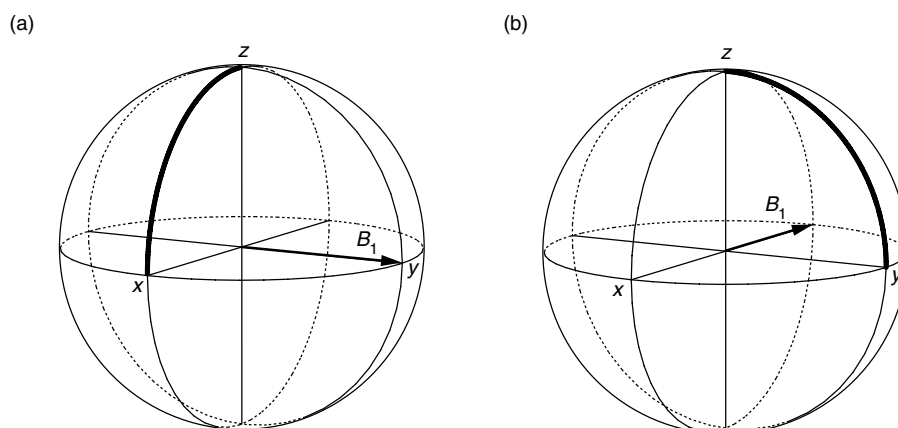


Fig. 3.22 Grapefruit plots showing the effect on equilibrium magnetization of (a) a 90° pulse about the y axis and (b) a 90° pulse about the $-x$ axis. Note the position of the B_1 field in each case.

x -pulse to have a phase shift of zero); likewise a pulse about $-x$ would be said to be phase shifted by 180° . On modern spectrometers it is possible to produce pulses with arbitrary phase shifts.

If we apply a 90° pulse about the y axis to equilibrium magnetization we find that the vector rotates in the yz -plane such that the magnetization ends up along x ; this is illustrated in Fig. 3.22. As before, we can determine the effect of such a pulse by thinking of it as causing a positive rotation about the y axis. A 90° pulse about $-x$ causes the magnetization to appear along y , as is also shown in Fig. 3.22.

We have seen that a 180° pulse about the x axis causes the vectors to move to mirror image positions with respect to the xz -plane. In a similar way,

a 180° pulse about the y axis causes the vectors to be reflected in the yz -plane.

3.10 Relaxation

We will have a lot more to say about relaxation later on, but at this point we will just note that the magnetization has a tendency to return to its equilibrium position (and size) – a process known as *relaxation*. Recall that the equilibrium situation has magnetization of size M_0 along z and no transverse (x or y) magnetization.

So, if we have created some transverse magnetization (for example by applying a 90° pulse) over time relaxation will cause this magnetization to decay away to zero. The free induction signal, which results from the magnetization precessing in the xy plane will therefore decay away in amplitude. This loss of x - and y -magnetization is called *transverse relaxation*.

Once perturbed, the z -magnetization will try to return to its equilibrium position, and this process is called *longitudinal relaxation*. We can measure the rate of this process using the *inversion recovery* experiment whose pulse sequence is shown in Fig. 3.23. The 180° pulse rotates the equilibrium magnetization to the $-z$ axis. Suppose that the delay τ is very short so that at the end of this delay the magnetization has not changed. Now the 90° pulse will rotate the magnetization onto the $+y$ axis; note that this is in contrast to the case where the magnetization starts out along $+z$ and it is rotated onto $-y$. If this gives a positive line in the spectrum, then having the magnetization along $+y$ will give a negative line. So, what we see for short values of τ is a negative line.

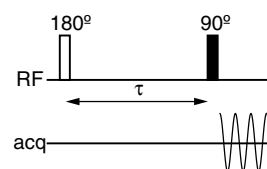


Fig. 3.23 The pulse sequence for the inversion recovery experiment used to measure longitudinal relaxation.

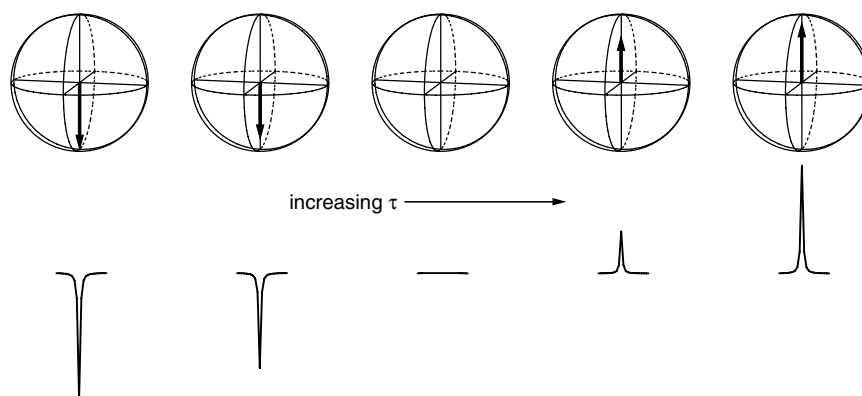


Fig. 3.24 Visualization of the outcome of an inversion recovery experiment. The size and sign of the z -magnetization is reflected in the spectra (shown underneath). By analysing the peak heights as a function of the delay τ it is possible to find the rate of recovery of the z -magnetization.

As τ gets longer more relaxation takes place and the magnetization shrinks towards zero; this result is a negative line in the spectrum, but one whose size is decreasing. Eventually the magnetization goes through zero and then starts to increase along $+z$ – this gives a positive line in the spectrum. Thus, by recording spectra with different values of the delay τ we can map out the recovery of the z -magnetization from the intensity of the observed lines. The

whole process is visualized in Fig. 3.24.

3.11 Off-resonance effects and soft pulses

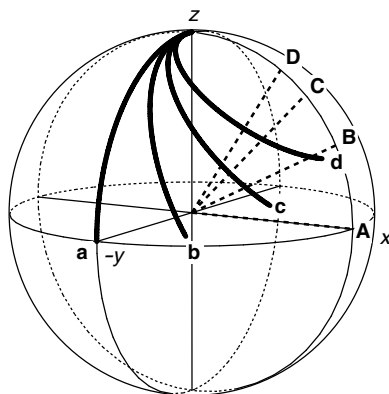


Fig. 3.25 Grapefruit diagram showing the path followed during a pulse for various different resonance offsets. Path **a** is for the on-resonance case; the effective field lies along x and is indicated by the dashed line **A**. Path **b** is for the case where the offset is half the RF field strength; the effective field is marked **B**. Paths **c** and **d** are for offsets equal to and 1.5 times the RF field strength, respectively. The effective field directions are labelled **C** and **D**.

So far we have only dealt with the case where the pulse is either on resonance or where the RF field strength is large compared to the offset (a hard pulse) which is in effect the same situation. We now turn to the case where the offset is comparable to the RF field strength. The consequences of this are sometimes a problem to us, but they can also be turned to our advantage for *selective excitation*.

As the offset becomes comparable to the RF field, the effective field begins to move up from the x axis towards the z axis. As a consequence, rather than the magnetization moving in the yz plane from z to $-y$, the magnetization starts to follow a more complex curved path; this is illustrated in Fig. 3.25. The further off resonance we go, the further the vector ends up from the xy plane. Also, there is a significant component of magnetization generated along the x direction, something which does not occur in the on-resonance case.

We can see more clearly what is going on if we plot the x and y magnetization as a function of the offset; these are shown in Fig. 3.26. In (a) we see the y -magnetization and, as expected for a 90° pulse, on resonance the equilibrium magnetization ends up entirely along $-y$. However, as the offset increases the amount of y -magnetization generally decreases but imposed on this overall decrease there is an oscillation; at some offsets the magnetization is zero and at others it is positive. The plot of the x -magnetization, (b), shows a similar story with the magnetization generally falling off as the offset increases, but again with a strong oscillation.

Plot (c) is of the magnitude of the magnetization, which is given by

$$M_{\text{abs}} = \sqrt{M_x^2 + M_y^2}.$$

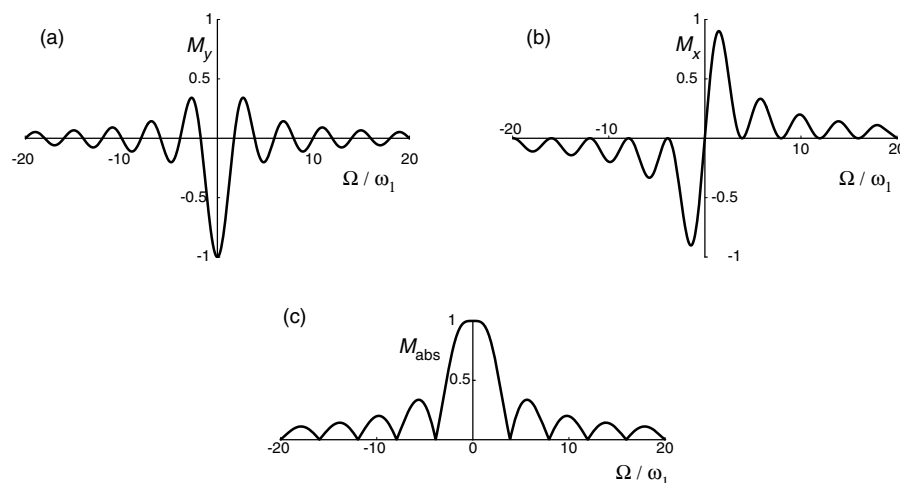


Fig. 3.26 Plots of the magnetization produced by a pulse as a function of the offset. The pulse length has been adjusted so that on resonance the flip angle is 90° . The horizontal axes of the plots is the offset expressed as a ratio of the RF field strength, ω_1 ; the equilibrium magnetization has been assumed to be of size 1.

This gives the total transverse magnetization in any direction; it is, of course, always positive. We see from this plot the characteristic nulls and subsidiary maxima as the offset increases.

What plot (c) tells us is that although a pulse can excite magnetization over a wide range of offsets, the region over which it does so efficiently is really rather small. If we want at least 90% of the full intensity the offset must be less than about 1.6 times the RF field strength.

Excitation of a range of shifts

There are some immediate practical consequences of this observation. Suppose that we are trying to record the full range of carbon-13 shifts (200 ppm) on an spectrometer whose magnetic field gives a proton Larmor frequency of 800 MHz and hence a carbon-13 Larmor frequency of 200 MHz. If we place the transmitter frequency at 100 ppm, the maximum offset that a peak can have is 100 ppm which, at this Larmor frequency, translates to 20 kHz. According to our criterion above, if we accept a reduction to 90% of the full intensity at the edges of the spectrum we would need an RF field strength of $20/1.6 \approx 12.5$ kHz. This would correspond to a 90° pulse width of $20 \mu\text{s}$. If the manufacturer failed to provide sufficient power to produce this pulse width we can see that the excitation of the spectrum will fall below our (arbitrary) 90% mark.

We will see in a later section that the presence of a mixture of x - and y -magnetization leads to phase errors in the spectrum which can be difficult to correct.

Selective excitation

Sometimes we want to excite just a portion of the spectrum, for example just the lines of a single multiplet. We can achieve this by putting the transmitter in the centre of the region we wish to excite and then reducing the RF field

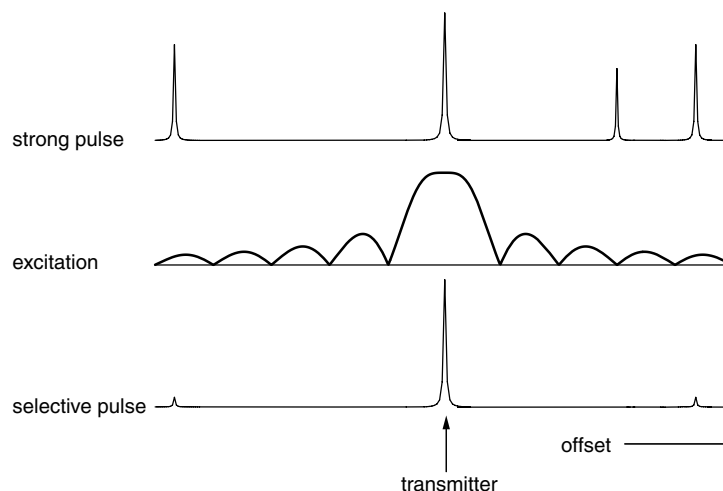


Fig. 3.27 Visualization of the use of selective excitation to excite just one line in the spectrum. At the top is shown the spectrum that would be excited using a hard pulse. If the transmitter is placed on resonance with one line and the strength of the RF field reduced then the pattern of excitation we expect is as shown in the middle. As a result, the peaks at non-zero offsets are attenuated and the spectrum which is excited will be as shown at the bottom.

strength until the degree of excitation of the rest of the spectrum is essentially negligible. Of course, reducing the RF field strength lengthens the duration of a 90° pulse. The whole process is visualized in Fig. 3.27.

Such pulses which are designed to affect only part of the spectrum are called *selective pulses* or *soft pulses* (as opposed to non-selective or hard pulses). The value to which we need to reduce the RF field depends on the separation of the peak we want to excite from those we do not want to excite. The closer in the unwanted peaks are the weaker the RF field must become and hence the longer the 90° pulse. In the end a balance has to be made between making the pulse too long (and hence losing signal due to relaxation) and allowing a small amount of excitation of the unwanted signals.

Figure 3.27 does not portray one problem with this approach, which is that for peaks away from the transmitter a mixture of x - and y -magnetization is generated (as shown in Fig. 3.26). This is described as a phase error, more of which in a later section. The second problem that the figure does show is that the excitation only falls off rather slowly and “bounces” through a series of maximum and nulls; these are sometimes called “wiggles”. We might be lucky and have an unwanted peak fall on a null, or unlucky and have an unwanted peak fall on a maximum.

Much effort has been put into getting round both of these problems. The key feature of all of the approaches is to “shape” the envelope of the RF pulses i.e. not just switch it on and off abruptly, but with a smooth variation. Such pulses are called *shaped pulses*. The simplest of these are basically bell-shaped (like a gaussian function, for example). These suppress the “wiggles” at large offsets and give just a smooth decay; they do not, however, improve the phase properties. To attack this part of the problem requires an altogether more sophisticated approach.

Selective inversion

Sometimes we want to invert the magnetization associated with just one resonance while leaving all the others in the spectrum unaffected; such a pulse would be called a *selective inversion* pulse. Just as for selective excitation, all we need to do is to place the transmitter on the line we wish to invert and reduce the RF field until the other resonances in the spectrum are not affected significantly. Of course we need to adjust the pulse duration so that the on-resonance flip angle is 180° .

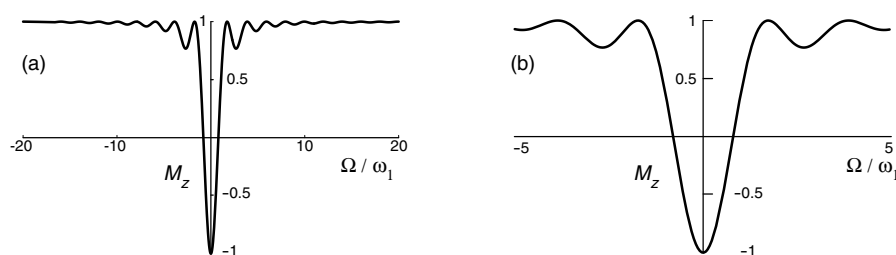


Fig. 3.28 Plots of the z -magnetization produced by a pulse as a function of the offset; the flip angle on-resonance has been set to 180° . Plot (b) covers a narrower range of offsets than plot (a). These plots should be compared with those in Fig. 3.26.

Figure 3.28 shows the z -magnetization generated as a function of offset for a pulse. We see that the range over which there is significant inversion is rather small, and that the oscillations are smaller in amplitude than for the excitation pulse.

This observation has two consequences: one “good” and one “bad”. The good consequence is that a selective 180° pulse is, for a given field strength, more selective than a corresponding 90° pulse. In particular, the weaker “bouncing” sidelobes are a useful feature. Do not forget, though, that the 180° pulse is longer, so some of the improvement may be illusory!

The bad consequence is that when it comes to hard pulses the range of offsets over which there is anything like complete inversion is much more limited than the range of offsets over which there is significant excitation. This can be seen clearly by comparing Fig. 3.28 with Figure 3.26. Thus, 180° pulses are often the source of problems in spectra with large offset ranges.

3.12 Exercises

E 3-1

A spectrometer operates with a Larmor frequency of 600 MHz for protons. For a particular set up the RF field strength, $\omega_1/(2\pi)$ has been determined to be 25 kHz. Suppose that the transmitter is placed at 5 ppm; what is the offset (in Hz) of a peak at 10 ppm? Compute the tilt angle, θ , of a spin with this offset.

For the normal range of proton shifts (0 – 10 ppm), is this 25 kHz field strong enough to give what could be classed as hard pulses?

E 3-2

In an experiment to determine the pulse length an operator observed a positive signal for pulse widths of 5 and 10 μs ; as the pulse was lengthened further the intensity decreased going through a null at 20.5 μs and then turning negative.

Explain what is happening in this experiment and use the data to determine the RF field strength in Hz and the length of a 90° pulse.

A further null in the signal was seen at 41.0 μs ; to what do you attribute this?

E 3-3

Use vector diagrams to describe what happens during a spin echo sequence in which the 180° pulse is applied about the y axis. Also, draw a phase evolution diagram appropriate for this pulse sequence.

In what way is the outcome different from the case where the refocusing pulse is applied about the x axis?

What would the effect of applying the refocusing pulse about the $-x$ axis be?

E 3-4

The gyromagnetic ratio of phosphorus-31 is $1.08 \times 10^8 \text{ rad s}^{-1} \text{ T}^{-1}$. This nucleus shows a wide range of shifts, covering some 700 ppm.

Estimate the minimum 90° pulse width you would need to excite peaks in this complete range to within 90% of their theoretical maximum for a spectrometer with a B_0 field strength of 9.4 T.

E 3-5

A spectrometer operates at a Larmor frequency of 400 MHz for protons and hence 100 MHz for carbon-13. Suppose that a 90° pulse of length 10 μs is applied to the protons. Does this have a significant effect of the carbon-13 nuclei? Explain your answer carefully.

E 3-6

Referring to the plots of Fig. 3.26 we see that there are some offsets at which the transverse magnetization goes to zero. Recall that the magnetization is rotating about the *effective field*, ω_{eff} ; it follows that these nulls in the excitation

come about when the magnetization executes complete 360° rotations about the effective field. In such a rotation the magnetization is returned to the z axis. Make a sketch of a “grapefruit” showing this.

The effective field is given by

$$\omega_{\text{eff}} = \sqrt{\omega_1^2 + \Omega^2}.$$

Suppose that we express the offset as a multiple κ of the RF field strength:

$$\Omega = \kappa \omega_1.$$

Show that with this values of Ω the effective field is given by:

$$\omega_{\text{eff}} = \omega_1 \sqrt{1 + \kappa^2}.$$

(The reason for doing this is to reduce the number of variables.)

Let us assume that on-resonance the pulse flip angle is $\pi/2$, so the duration of the pulse, τ_p , is give from

$$\omega_1 \tau_p = \pi/2 \quad \text{thus} \quad \tau_p = \frac{\pi}{2\omega_1}.$$

The angle of rotation about the effective field for a pulse of duration τ_p is $(\omega_{\text{eff}} \tau_p)$. Show that for the effective field given above this angle, β_{eff} is given by

$$\beta_{\text{eff}} = \frac{\pi}{2} \sqrt{1 + \kappa^2}.$$

The null in the excitation will occur when β_{eff} is 2π i.e. a complete rotation. Show that this occurs when $\kappa = \sqrt{15}$ i.e. when $(\Omega/\omega_1) = \sqrt{15}$. Does this agree with Fig. 3.26?

Predict other values of κ at which there will be nulls in the excitation.

E 3-7

When calibrating a pulse by looking for the null produced by a 180° rotation, why is it important to choose a line which is close to the transmitter frequency (i.e. one with a small offset)?

E 3-8

Use vector diagrams to predict the outcome of the sequence:

$$90^\circ - \text{delay } \tau - 90^\circ$$

applied to equilibrium magnetization; both pulses are about the x axis. In your answer, explain how the x , y and z magnetizations depend on the delay τ and the offset Ω .

E 3-9

Consider the spin echo sequence to which a 90° pulse has been added at the end:

$$90^\circ(x) - \text{delay } \tau - 180^\circ(x) - \text{delay } \tau - 90^\circ(\phi).$$

The axis about which the pulse is applied is given in brackets after the flip angle. Explain in what way the outcome is different depending on whether the phase ϕ of the pulse is chosen to be x , y , $-x$ or $-y$.

E 3-10

The so-called “1- $\bar{1}$ ” sequence is:

$$90^\circ(x) - \text{delay } \tau - 90^\circ(-x)$$

For a peak which is on resonance the sequence does not excite any observable magnetization. However, for a peak with an offset such that $\Omega\tau = \pi/2$ the sequence results in all of the equilibrium magnetization appearing along the x axis. Further, if the delay is such that $\Omega\tau = \pi$ no transverse magnetization is excited.

Explain these observations and make a sketch graph of the amount of transverse magnetization generated as a function of the offset for a fixed delay τ .

The sequence has been used for suppressing strong solvent signals which might otherwise overwhelm the spectrum. The solvent is placed on resonance, and so is not excited; τ is chosen so that the peaks of interest are excited. How does one go about choosing the value for τ ?

E 3-11

The so-called “1-1” sequence is:

$$90^\circ(x) - \text{delay } \tau - 90^\circ(y).$$

Describe the excitation that this sequence produces as a function of offset. How it could be used for observing spectra in the presence of strong solvent signals?

4 Fourier transformation and data processing

In the previous chapter we have seen how the precessing magnetization can be detected to give a signal which oscillates at the Larmor frequency – the free induction signal. We also commented that this signal will eventually decay away due to the action of relaxation; the signal is therefore often called the *free induction decay* or FID. The question is how do we turn this signal, which depends on *time*, into the a spectrum, in which the horizontal axis is *frequency*.

This conversion is made using a mathematical process known as *Fourier transformation*. This process takes the *time domain* function (the FID) and converts it into a *frequency domain* function (the spectrum); this is shown in Fig. 4.1. In this chapter we will start out by exploring some features of the spectrum, such as phase and lineshapes, which are closely associated with the Fourier transform and then go on to explore some useful manipulations of NMR data such as sensitivity and resolution enhancement.

4.1 The FID

In section 3.6 we saw that the x and y components of the free induction signal could be computed by thinking about the evolution of the magnetization during the acquisition time. In that discussion we assumed that the magnetization started out along the $-y$ axis as this is where it would be rotated to by a 90° pulse. For the purposes of this chapter we are going to assume that the magnetization starts out along x ; we will see later that this choice of starting position is essentially arbitrary.

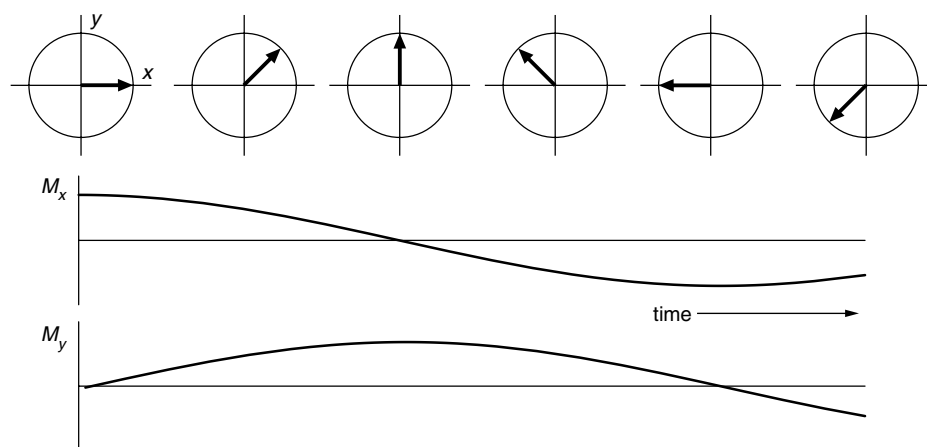


Fig. 4.2 Evolution of the magnetization over time; the offset is assumed to be positive and the magnetization starts out along the x axis.

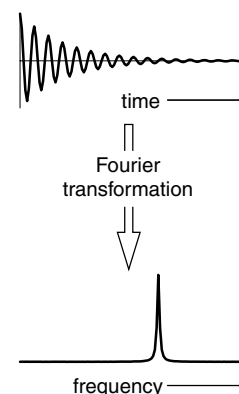


Fig. 4.1 Fourier transformation is the mathematical process which takes us from a function of time (the time domain) – such as a FID – to a function of frequency – the spectrum.

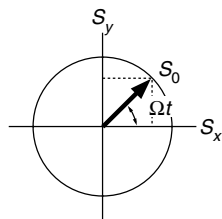


Fig. 4.3 The x and y components of the signal can be thought of as arising from the rotation of a vector S_0 at frequency Ω .

If the magnetization does indeed start along x then Fig. 3.16 needs to be redrawn, as is shown in Fig. 4.2. From this we can easily see that the x and y components of the magnetization are:

$$M_x = M_0 \cos \Omega t$$

$$M_y = M_0 \sin \Omega t.$$

The signal that we detect is proportional to these magnetizations. The constant of proportion depends on all sorts of instrumental factors which need not concern us here; we will simply write the detected x and y signals, $S_x(t)$ and $S_y(t)$ as

$$S_x(t) = S_0 \cos \Omega t \quad \text{and} \quad S_y(t) = S_0 \sin \Omega t$$

where S_0 gives the overall size of the signal and we have reminded ourselves that the signal is a function of time by writing it as $S_x(t)$ etc.

It is convenient to think of this signal as arising from a vector of length S_0 rotating at frequency Ω ; the x and y components of the vector give S_x and S_y , as is illustrated in Fig. 4.3.

As a consequence of the way the Fourier transform works, it is also convenient to regard $S_x(t)$ and $S_y(t)$ as the real and imaginary parts of a complex signal $S(t)$:

$$\begin{aligned} S(t) &= S_x(t) + i S_y(t) \\ &= S_0 \cos \Omega t + i S_0 \sin \Omega t \\ &= S_0 \exp(i \Omega t). \end{aligned}$$

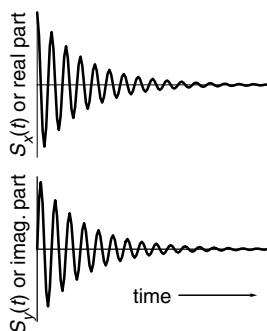


Fig. 4.4 Illustration of a typical FID, showing the real and imaginary parts of the signal; both decay over time.

We need not concern ourselves too much with the mathematical details here, but just note that the time-domain signal is complex, with the real and imaginary parts corresponding to the x and y components of the signal.

We mentioned at the start of this section that the transverse magnetization decays over time, and this is most simply represented by an exponential decay with a time constant T_2 . The signal then becomes

$$S(t) = S_0 \exp(i \Omega t) \exp\left(\frac{-t}{T_2}\right). \quad (4.1)$$

A typical example is illustrated in Fig. 4.4. Another way of writing this is to define a (first order) rate constant $R_2 = 1/T_2$ and so $S(t)$ becomes

$$S(t) = S_0 \exp(i \Omega t) \exp(-R_2 t). \quad (4.2)$$

The shorter the time T_2 (or the larger the rate constant R_2) the more rapidly the signal decays.

4.2 Fourier transformation

Fourier transformation of a signal such as that given in Eq. 4.1 gives the frequency domain signal which we know as the spectrum. Like the time domain signal the frequency domain signal has a real and an imaginary part. The real

part of the spectrum shows what we call an *absorption mode* line, in fact in the case of the exponentially decaying signal of Eq. 4.1 the line has a shape known as a *Lorentzian*, or to be precise the *absorption mode Lorentzian*. The imaginary part of the spectrum gives a lineshape known as the *dispersion mode Lorentzian*. Both lineshapes are illustrated in Fig. 4.5.

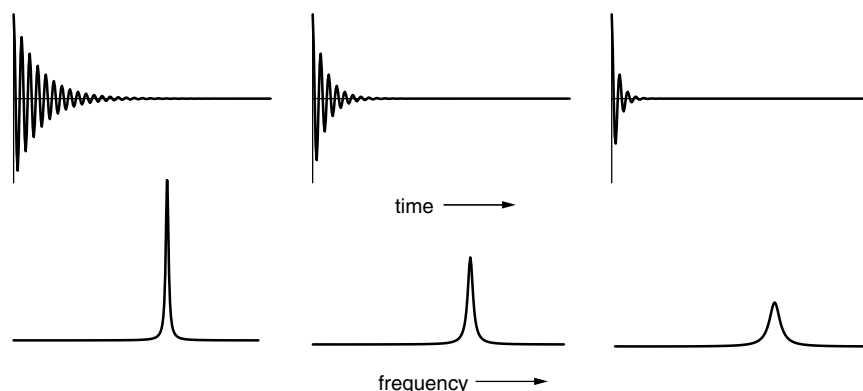


Fig. 4.6 Illustration of the fact that the more rapidly the FID decays the broader the line in the corresponding spectrum. A series of FIDs are shown at the top of the figure and below are the corresponding spectra, all plotted on the same vertical scale. The integral of the peaks remains constant, so as they get broader the peak height decreases.

This absorption lineshape has a width at half of its maximum height of $1/(\pi T_2)$ Hz or (R/π) Hz. This means that the faster the decay of the FID the broader the line becomes. However, the area under the line – that is the integral – remains constant so as it gets broader so the peak height reduces; these points are illustrated in Fig. 4.6.

If the size of the time domain signal increases, for example by increasing S_0 the height of the peak increases in direct proportion. These observations lead to the very important consequence that by integrating the lines in the spectrum we can determine the relative number of protons (typically) which contribute to each.

The dispersion line shape is not one that we would choose to use. Not only is it broader than the absorption mode, but it also has positive and negative parts. In a complex spectrum these might cancel one another out, leading to a great deal of confusion. If you are familiar with ESR spectra you might recognize the dispersion mode lineshape as looking like the *derivative* lineshape which is traditionally used to plot ESR spectra. Although these two lineshapes do look roughly the same, they are not in fact related to one another.

Positive and negative frequencies

As we discussed in section 3.5, the evolution we observe is at frequency Ω i.e. the apparent Larmor frequency in the rotating frame. This offset can be positive or negative and, as we will see later, it turns out to be possible to determine the sign of the frequency. So, in our spectrum we have positive and negative frequencies, and it is usual to plot these with zero in the middle.

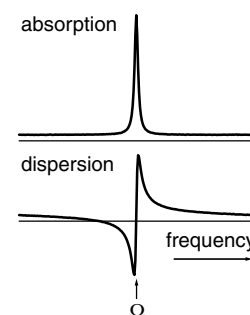


Fig. 4.5 Illustration of the absorption and dispersion mode Lorentzian lineshapes. Whereas the absorption lineshape is always positive, the dispersion lineshape has positive and negative parts; it also extends further.

Several lines

What happens if we have more than one line in the spectrum? In this case, as we saw in section 3.5, the FID will be the sum of contributions from each line. For example, if there are three lines $S(t)$ will be:

$$S(t) = S_{0,1} \exp(i\Omega_1 t) \exp\left(\frac{-t}{T_2^{(1)}}\right) + S_{0,2} \exp(i\Omega_2 t) \exp\left(\frac{-t}{T_2^{(2)}}\right) + S_{0,3} \exp(i\Omega_3 t) \exp\left(\frac{-t}{T_2^{(3)}}\right).$$

where we have allowed each line to have a separate intensity, $S_{0,i}$, frequency, Ω_i , and relaxation time constant, $T_2^{(i)}$.

The Fourier transform is a *linear* process which means that if the time domain is a sum of functions the frequency domain will be a sum of Fourier transforms of those functions. So, as Fourier transformation of each of the terms in $S(t)$ gives a line of appropriate width and frequency, the Fourier transformation of $S(t)$ will be the sum of these lines – which is the complete spectrum, just as we require it.

4.3 Phase

So far we have assumed that at time zero (i.e. at the start of the FID) $S_x(t)$ is a maximum and $S_y(t)$ is zero. However, in general this need not be the case – it might just as well be the other way round or anywhere in between. We describe this general situation by saying that the signal is *phase shifted* or that it has a *phase error*. The situation is portrayed in Fig. 4.7.

In Fig. 4.7 (a) we see the situation we had before, with the signal starting out along x and precessing towards y . The real part of the FID (corresponding to S_x) is a damped cosine wave and the imaginary part (corresponding to S_y) is a damped sine wave. Fourier transformation gives a spectrum in which the real part contains the absorption mode lineshape and the imaginary part the dispersion mode.

In (b) we see the effect of a phase shift, ϕ , of 45° . S_y now starts out at a finite value, rather than at zero. As a result neither the real nor the imaginary part of the spectrum has the absorption mode lineshape; both are a mixture of absorption and dispersion.

In (c) the phase shift is 90° . Now it is S_y which takes the form of a damped cosine wave, whereas S_x is a sine wave. The Fourier transform gives a spectrum in which the absorption mode signal now appears in the imaginary part. Finally in (d) the phase shift is 180° and this gives a negative absorption mode signal in the real part of the spectrum.

What we see is that in general the appearance of the spectrum depends on the position of the signal at time zero, that is on the phase of the signal at time zero. Mathematically, inclusion of this phase shift means that the (complex) signal becomes:

$$S(t) = S_0 \exp(i\phi) \exp(i\Omega t) \exp\left(\frac{-t}{T_2}\right). \quad (4.3)$$

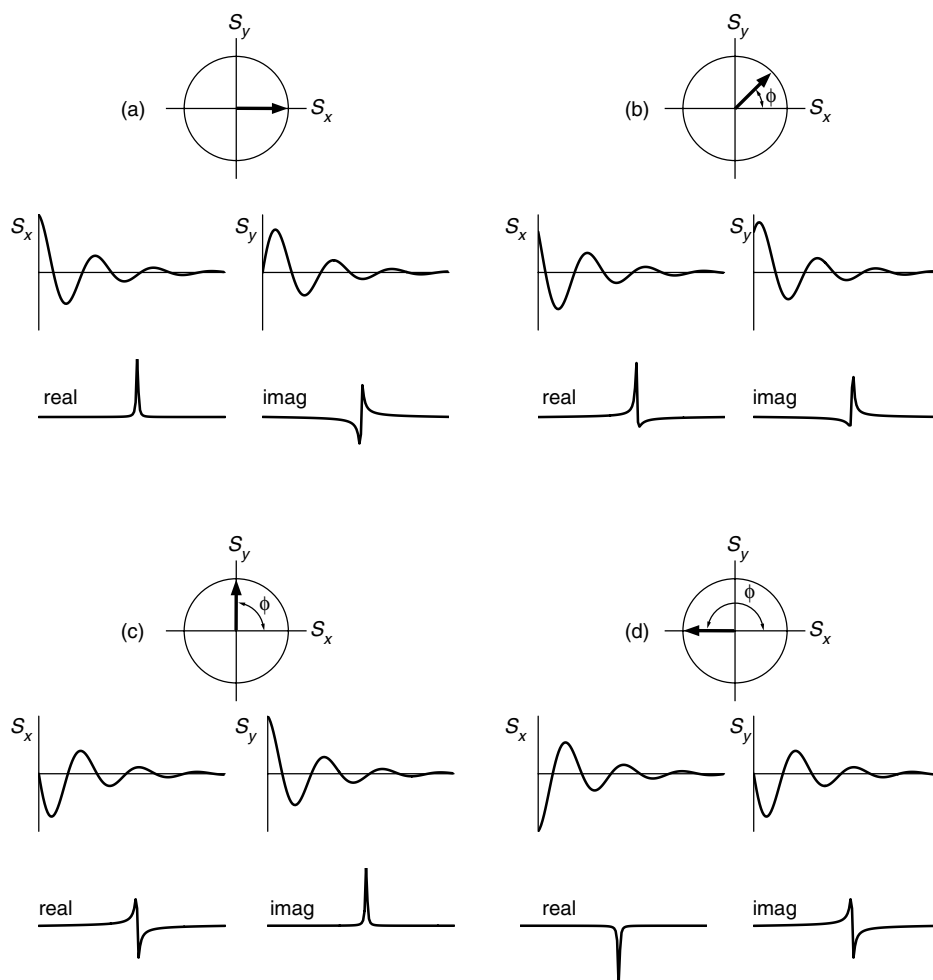


Fig. 4.7 Illustration of the effect of a phase shift of the time domain signal on the spectrum. In (a) the signal starts out along x and so the spectrum is the absorption mode in the real part and the dispersion mode in the imaginary part. In (b) there is a phase shift, ϕ , of 45° ; the real and imaginary parts of the spectrum are now mixtures of absorption and dispersion. In (c) the phase shift is 90° ; now the absorption mode appears in the imaginary part of the spectrum. Finally in (d) the phase shift is 180° giving a negative absorption line in the real part of the spectrum. The vector diagrams illustrate the position of the signal at time zero.

Phase correction

It turns out that for instrumental reasons the axis along which the signal appears cannot be predicted, so in any practical situation there is an unknown phase shift. In general, this leads to a situation in which the real part of the spectrum (which is normally the part we display) does not show a pure absorption lineshape. This is undesirable as for the best resolution we require an absorption mode lineshape.

Luckily, restoring the spectrum to the absorption mode is easy. Suppose we take the FID, represented by Eq. 4.3, and multiply it by $\exp(i\phi_{\text{corr}})$:

$$\exp(i\phi_{\text{corr}})S(t) = \exp(i\phi_{\text{corr}}) \times \left[S_0 \exp(i\phi) \exp(i\Omega t) \exp\left(\frac{-t}{T_2}\right) \right].$$

This is easy to do as by now the FID is stored in computer memory, so the

multiplication is just a mathematical operation on some numbers. Exponentials have the property that $\exp(A)\exp(B) = \exp(A + B)$ so we can re-write the time domain signal as

$$\exp(i\phi_{\text{corr}})S(t) = \exp(i(\phi_{\text{corr}} + \phi)) \left[S_0 \exp(i\Omega t) \exp\left(\frac{-t}{T_2}\right) \right].$$

Now suppose that we set $\phi_{\text{corr}} = -\phi$; as $\exp(0) = 1$ the time domain signal becomes:

$$\exp(i\phi_{\text{corr}})S(t) = S_0 \exp(i\Omega t) \exp\left(\frac{-t}{T_2}\right).$$

The signal now has no phase shift and so will give us a spectrum in which the real part shows the absorption mode lineshape – which is exactly what we want. All we need to do is find the correct ϕ_{corr} .

It turns out that the phase correction can just as easily be applied to the spectrum as it can to the FID. So, if the spectrum is represented by $S(\omega)$ (a function of frequency, ω) the phase correction is applied by computing

$$\exp(i\phi_{\text{corr}})S(\omega).$$

Such a correction is called a *frequency independent* or *zero order* phase correction as it is the same for all peaks in the spectrum, regardless of their offset.

In practice what happens is that we Fourier transform the FID and display the real part of the spectrum. We then adjust the phase correction (i.e. the value of ϕ_{corr}) until the spectrum appears to be in the absorption mode – usually this adjustment is made by turning a knob or by a “click and drag” operation with the mouse. The whole process is called *phasing the spectrum* and is something we have to do each time we record a spectrum.

In addition to the phase shifts introduced by the spectrometer we can of course deliberately introduce a shift of phase by, for example, altering the phase of a pulse. In a sense it does not matter what the phase of the signal is – we can always obtain an absorption spectrum by phase correcting the spectrum later on.

Frequency dependent phase errors

We saw in section 3.11 that if the offset becomes comparable with the RF field strength a 90° pulse about x results in the generation of magnetization along both the x and y axes. This is in contrast to the case of a hard pulse, where the magnetization appears only along $-y$. We can now describe this mixture of x and y magnetization as resulting in a *phase shift* or *phase error* of the spectrum.

Figure 3.25 illustrates very clearly how the x component increases as the offset increases, resulting in a phase error which also increases with offset. Therefore lines at different offsets in the spectrum will have different phase errors, the error increasing as the offset increases. This is illustrated schematically in the upper spectrum shown in Fig. 4.8.

If there were only one line in the spectrum it would be possible to ensure that the line appeared in the absorption mode simply by adjusting the phase

Attempts have been made over the years to automate this phasing process; on well resolved spectra the results are usually good, but these automatic algorithms tend to have more trouble with poorly-resolved spectra. In any case, what constitutes a correctly phased spectrum is rather subjective.

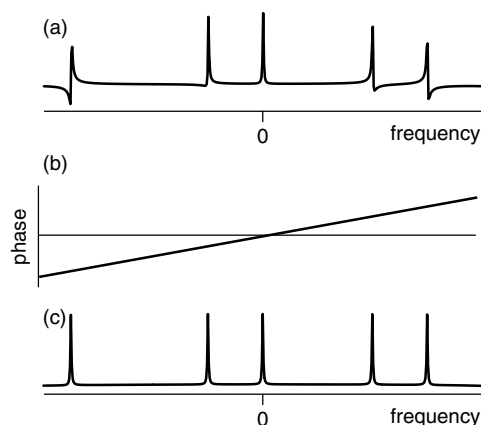


Fig. 4.8 Illustration of the appearance of a frequency dependent phase error in the spectrum. In (a) the line which is on resonance (at zero frequency) is in pure absorption, but as the offset increases the phase error increases. Such an frequency dependent phase error would result from the use of a pulse whose RF field strength was not much larger than the range of offsets. The spectrum can be returned to the absorption mode, (c), by applying a phase correction which varies with the offset in a linear manner, as shown in (b). Of course, to obtain a correctly phased spectrum we have to choose the correct slope of the graph of phase against offset.

in the way described above. However, if there is more than one line present in the spectrum the phase correction for each will be *different*, and so it will be impossible to phase all of the lines at once.

Luckily, it is often the case that the phase correction needed is directly proportional to the offset – called a *linear* or *first order* phase correction. Such a variation in phase with offset is shown in Fig. 4.8 (b). All we have to do is to vary the rate of change of phase with frequency (the slope of the line) until the spectrum appears to be phased; as with the zero-order phase correction the computer software usually makes it easy for us to do this by turning a knob or pushing the mouse. In practice, to phase the spectrum correctly usually requires some iteration of the zero- and first-order phase corrections.

The usual convention is to express the frequency dependent phase correction as the value that the phase takes at the extreme edges of the spectrum. So, for example, such a correction by 100° means that the phase correction is zero in the middle (at zero offset) and rises linearly to $+100^\circ$ at one edge and falls linearly to -100° at the opposite edge.

For a pulse the phase error due to these off-resonance effects for a peak with offset Ω is of the order of (Ωt_p) , where t_p is the length of the pulse. For a carbon-13 spectrum recorded at a Larmor frequency of 125 MHz, the maximum offset is about 100 ppm which translates to 12500 Hz. Let us suppose that the 90° pulse width is $15 \mu\text{s}$, then the phase error is

$$2\pi \times 12500 \times 15 \times 10^{-6} \approx 1.2 \text{ radians}$$

which is about 68° ; note that in the calculation we had to convert the offset from Hz to rad s^{-1} by multiplying by 2π . So, we expect the frequency dependent phase error to vary from zero in the middle of the spectrum (where the offset is zero) to 68° at the edges; this is a significant effect.

For reasons which we cannot go into here it turns out that the linear phase

correction is sometimes only a first approximation to the actual correction needed. Provided that the lines in the spectrum are sharp a linear correction works very well, but for broad lines it is not so good. Attempting to use a first-order phase correction on such spectra often results in distortions of the baseline.

4.4 Sensitivity enhancement

Inevitably when we record a FID we also record noise at the same time. Some of the noise is contributed by the amplifiers and other electronics in the spectrometer, but the major contributor is the thermal noise from the coil used to detect the signal. Reducing the noise contributed by these two sources is largely a technical matter which will not concern use here. NMR is not a sensitive technique, so we need to take any steps we can to improve the signal-to-noise ratio in the spectrum. We will see that there are some manipulations we can perform on the FID which will give us some improvement in the signal-to-noise ratio (SNR).

By its very nature, the FID decays over time but in contrast the noise just goes on and on. Therefore, if we carry on recording data for long after the FID has decayed we will just measure noise and no signal. The resulting spectrum will therefore have a poor signal-to-noise ratio.

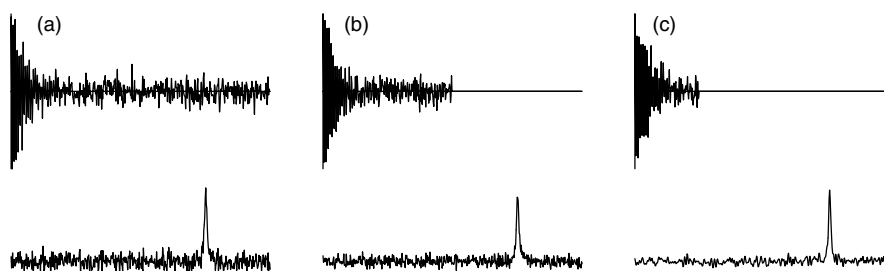


Fig. 4.9 Illustration of the effect of the time spent acquiring the FID on the signal-to-noise ratio (SNR) in the spectrum. In (a) the FID has decayed to next to nothing within the first quarter of the time, but the noise carries on unabated for the whole time. Shown in (b) is the effect of halving the time spent acquiring the data; the SNR improves significantly. In (c) we see that taking the first quarter of the data gives a further improvement in the SNR.

This point is illustrated in Fig. 4.9 where we see that by recording the FID for long after it has decayed all we end up doing is recording more noise and no signal. Just shortening the time spent recording the signal (called the *acquisition time*) will improve the SNR since more or less all the signal is contained in the early part of the FID. Of course, we must not shorten the acquisition time too much or we will start to miss the FID, which would result in a reduction in SNR.

Sensitivity enhancement

Looking at the FID we can see that at the start the signal is strongest. As time progresses, the signal decays and so gets weaker but the noise remains at the same level. The idea arises, therefore, that the early parts of the FID are

“more important” as it is here where the signal is the strongest.

This effect can be exploited by deliberately multiplying the FID by a function which starts at 1 and then steadily tails away to zero. The idea is that this function will cut off the later parts of the FID where the signal is weakest, but leave the early parts unaffected.

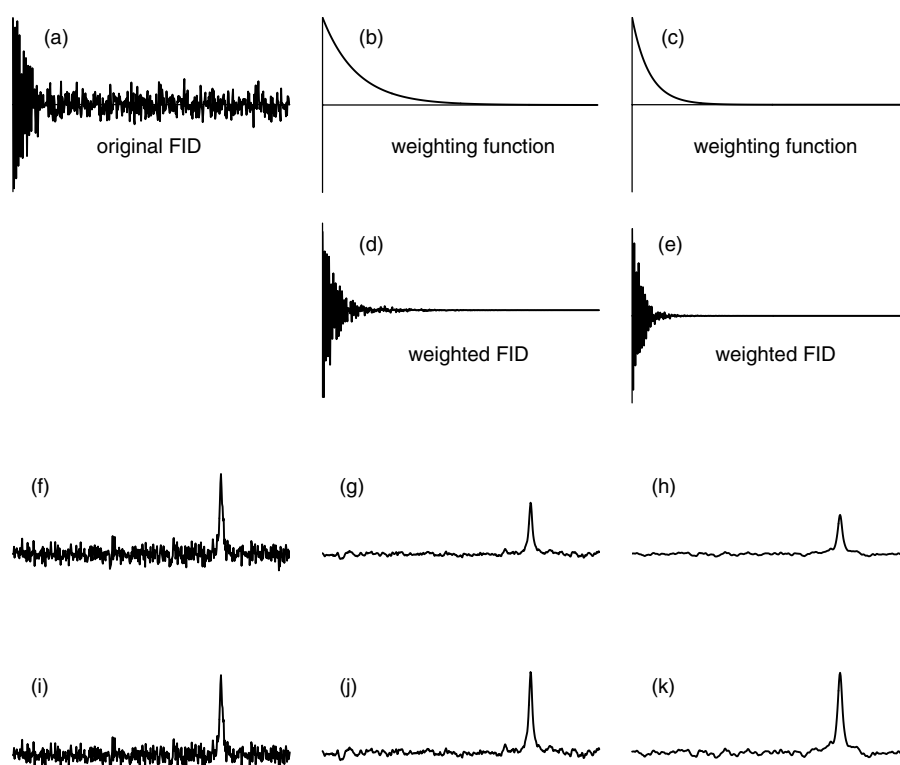


Fig. 4.10 Illustration of how multiplying a FID by a decaying function (a weighting function) can improve the SNR. The original FID is shown in (a) and the corresponding spectrum is (f). Multiplying the FID by a weighting function (b) gives (c); Fourier transformation of (c) gives the spectrum (g). Note the improvement in SNR of (g) compared to (f). Multiplying (a) by the more rapidly decaying weighting function (d) gives (e); the corresponding spectrum is (h). The improvement in SNR is less marked. Spectra (f) – (h) are all plotted on the same vertical scale so that the decrease in peak height can be seen. The same spectra are plotted in (i) – (k) but this time normalized so that the peak height is the same; this shows most clearly the improvement in the SNR.

A typical choice for this function – called a *weighting function* – is an exponential:

$$W(t) = \exp(-R_{LB}t) \quad (4.4)$$

where R_{LB} is a rate constant which we are free to choose. Figure 4.10 illustrates the effect on the SNR of different choices of this decay constant.

Spectrum (g) shows a large improvement in the SNR when compared to spectrum (f) simply because the long tail of noise is suppressed. Using a more rapidly decaying weighting function, (d), gives a further small improvement in the SNR (h). It should not be forgotten that the weighting function also acts on the signal, causing it to decay more quickly. As was explained in section 4.2 a more rapidly decaying signal leads to a broader line. So, the use

of the weighting function will not only attenuate the noise it will also broaden the lines; this is clear from Fig. 4.10.

Broadening the lines also reduces the peak height (remember that the integral remains constant) – something that is again evident from Fig. 4.10. Clearly, this reduction in peak height will reduce the SNR. Hence there is a trade-off we have to make: the more rapidly decaying the weighting function the more the noise in the tail of the FID is attenuated thus reducing the noise level in the spectrum. However, at the same time a more rapidly decaying function will cause greater line broadening, and this will reduce the SNR. It turns out that there is an optimum weighting function, called the *matched filter*.

Matched filter

Suppose that the FID can be represented by the exponentially decaying function introduced in Eq. 4.2:

$$S(t) = S_0 \exp(i\Omega t) \exp(-R_2 t).$$

The line in the corresponding spectrum is of width (R_2/π) Hz. Let us apply a weighting function of the kind described by Eq. 4.4 i.e. a decaying exponential:

$$W(t) \times S(t) = \exp(-R_{LB} t) [S_0 \exp(i\Omega t) \exp(-R_2 t)].$$

The decay due to the weighting function on its own would give a linewidth of (R_{LB}/π) Hz. Combining the two terms describing the exponential decays gives

$$W(t) \times S(t) = S_0 \exp(i\Omega t) \exp(-(R_{LB} + R_2)t).$$

From this we see that the weighted FID will give a linewidth of

$$(R_{LB} + R_2)/\pi.$$

In words, the linewidth in the spectrum is the sum of the linewidths in the original spectrum and the additional linebroadening imposed by the weighting function.

It is usual to specify the weighting function in terms of the extra line broadening it will cause. So, a “linebroadening of 1 Hz” is a function which will increase the linewidth in the spectrum by 1 Hz. For example if 5 Hz of linebroadening is required then $(R_{LB}/\pi) = 5$ giving $R_{LB} = 15.7 \text{ s}^{-1}$.

It can be shown that the best SNR is obtained by applying a weighting function which matches the linewidth in the original spectrum – such a weighting function is called a *matched filter*. So, for example, if the linewidth is 2 Hz in the original spectrum, applying an additional line broadening of 2 Hz will give the optimum SNR.

We can see easily from this argument that if there is a range of linewidths in the spectrum we cannot find a value of the linebroadening which is the optimum for all the peaks. Also, the extra line broadening caused by the matched filter may not be acceptable on the grounds of the decrease in resolution it causes. Under these circumstances we may choose to use sufficient line broadening to cut off the excess noise in the tail of the FID, but still less than the matched filter.

4.5 Resolution enhancement

We noted in section 4.2 that the more rapidly the time domain signal decays the broader the lines become. A weighting function designed to improve the SNR inevitably leads to a broadening of the lines as such a function hastens the decay of the signal. In this section we will consider the opposite case, where the weighting function is designed to *narrow* the lines in the spectrum and so increase the resolution.

The basic idea is simple. All we need to do is to multiply the FID by a weighting function which *increases* with time, for example a rising exponential:

$$W(t) = \exp(+R_{\text{RE}}t) \quad R_{\text{RE}} > 0.$$

This function starts at 1 when $t = 0$ and then rises indefinitely.

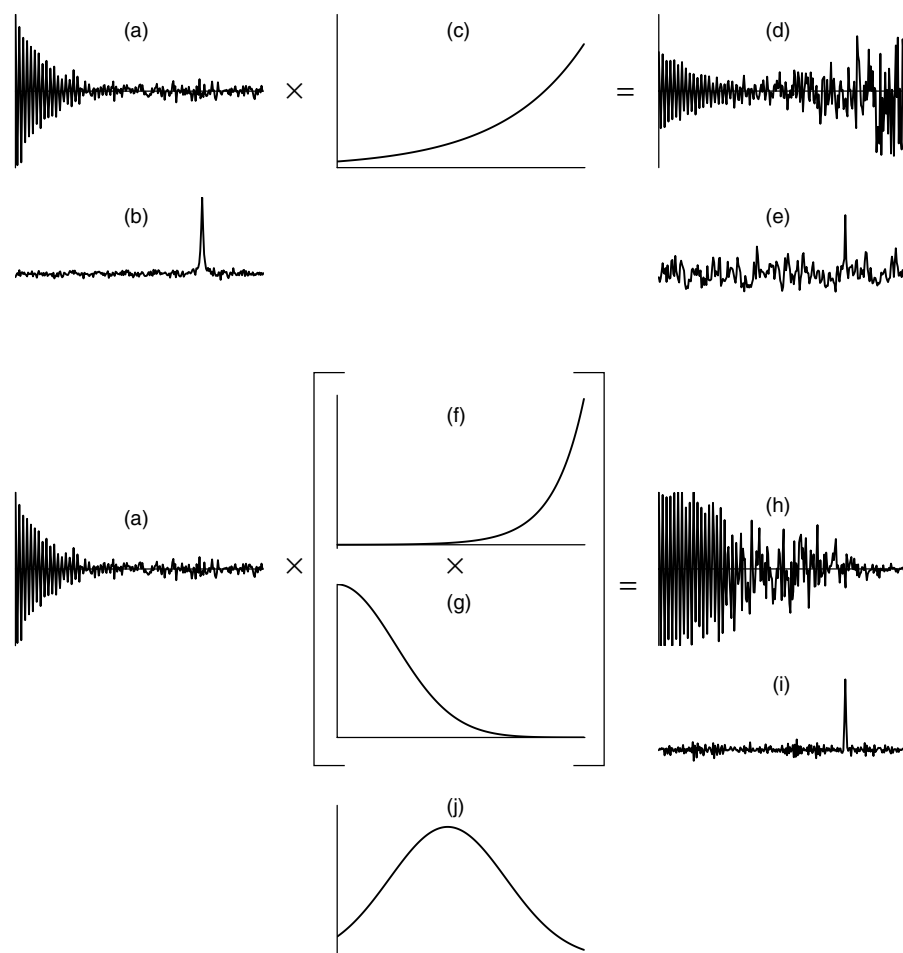


Fig. 4.11 Illustration of the use of weighting functions to enhance the resolution in the spectrum. Note that the scales of the plots have been altered to make the relevant features clear. See text for details.

The problem with multiplying the FID with such a function is that the noise in the tail of the FID is amplified, thus making the SNR in the spectrum very poor indeed. To get round this it is, after applying the positive exponential we multiply by a second decaying function to “clip” the noise at the

tail of the FID. Usually, the second function is chosen to be one that decays relatively slowly over most of the FID and then quite rapidly at the end – a common choice is the Gaussian function:

$$W(t) = \exp(-\alpha t^2),$$

where α is a parameter which sets the decay rate. The larger α , the faster the decay rate.

The whole process is illustrated in Fig. 4.11. The original FID, (a), contains a significant amount of noise and has been recorded well beyond the point where the signal decays into the noise. Fourier transformation of (a) gives the spectrum (b). If (a) is multiplied by the rising exponential function plotted in (c), the result is the FID (d); note how the decay of the signal has been slowed, but the noise in the tail of the FID has been greatly magnified. Fourier transformation of (d) gives the spectrum (e); the resolution has clearly been improved, but at the expense of a large reduction in the SNR.

Referring now to the bottom part of Fig. 4.11 we can see the effect of introducing a Gaussian weighting function as well. The original FID (a) is multiplied by the rising exponential (f) and the decaying Gaussian (g); this gives the time-domain signal (h). Note that once again the signal decay has been slowed, but the noise in the tail of the FID is not as large as it is in (d). Fourier transformation of (h) gives the spectrum (i); the resolution has clearly been improved when compared to (b), but without too great a loss of SNR.

Finally, plot (j) shows the product of the two weighting functions (f) and (g). We can see clearly from this plot how the two functions combine together to first increase the time-domain function and then to attenuate it at longer times. Careful choice of the parameters R_{RE} and α are needed to obtain the optimum result. Usually, a process of trial and error is adopted.

If we can set R_{RE} to cancel exactly the original decay of the FID then the result of this process is to generate a time-domain function which only has a Gaussian decay. The resulting peak in the spectrum will have a Gaussian lineshape, which is often considered to be superior to the Lorentzian as it is narrower at the base; the two lineshapes are compared in Fig. 4.12. This transformation to a Gaussian lineshape is often called the *Lorentz-to-Gauss* transformation.

Parameters for the Lorentz-to-Gauss transformation

The combined weighting function for this transformation is

$$W(t) = \exp(R_{RE}t) \exp(-\alpha t^2).$$

The usual approach is to specify R_{RE} in terms of the linewidth that it would create on its own if it were used to specify a *decaying* exponential. Recall that in such a situation the linewidth, L , is given by R_{RE}/π . Thus $R_{RE} = \pi L$. The weighting function can therefore be rewritten

$$W(t) = \exp(-\pi Lt) \exp(-\alpha t^2).$$

For compatibility with the linebroadening role of a decaying exponential it is usual to define the exponential weighting function as $\exp(-\pi Lt)$ so that

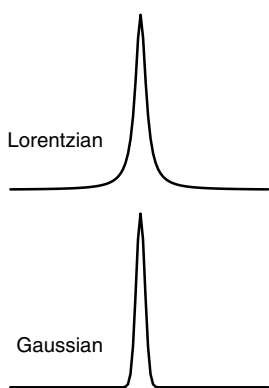


Fig. 4.12 Comparison of the Lorentzian and Gaussian lineshapes; the two peaks have been adjusted so that their peak heights and widths at half height are equal. The Gaussian is a more “compact” lineshape.

a positive value for L leads to line broadening and a negative value leads to resolution enhancement.

From Fig. 4.11 (j) we can see that the effect of the rising exponential and the Gaussian is to give an overall weighting function which has a maximum in it. It is usual to define the Gaussian parameter α from the position of this maximum. A little mathematics shows that this maximum occurs at time $t_{\max}$ given by:

$$t_{\max} = -\frac{L\pi}{2\alpha}$$

(recall that L is negative) so that

$$\alpha = -\frac{L\pi}{2t_{\max}}.$$

We simply select a value of $t_{\max}$ and use this to define the value of α . On some spectrometers, $t_{\max}$ is expressed as a fraction, f , of the acquisition time, t_{acq} : $t_{\max} = ft_{\text{acq}}$. In this case

$$\alpha = -\frac{L\pi}{2ft_{\text{acq}}}.$$

4.6 Other weighting functions

Many other weighting functions have been used for sensitivity enhancement and resolution enhancement. Perhaps the most popular are the *sine bell* variants on it, which are illustrated in Fig. 4.13.

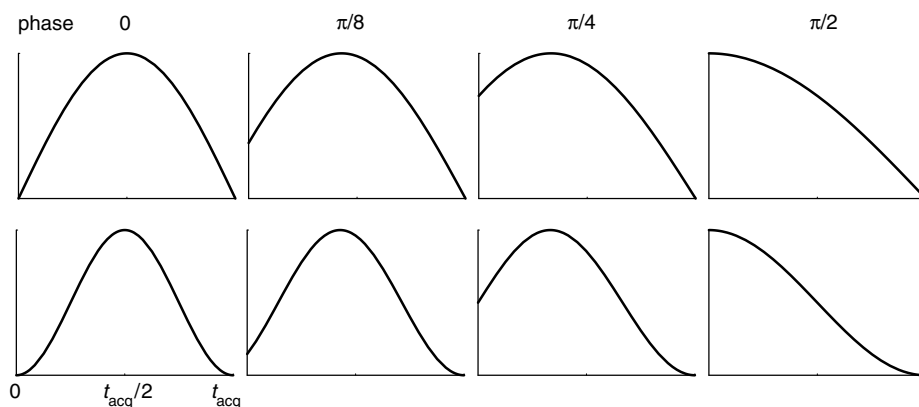


Fig. 4.13 The top row shows sine bell and the bottom row shows sine bell squared weighting functions for different choices of the phase parameter; see text for details.

The basic sine bell is just the first part of a $\sin \theta$ for $\theta = 0$ to $\theta = \pi$; this is illustrated in the top left-hand plot of Fig. 4.13. In this form the function will give resolution enhancement rather like the combination of a rising exponential and a Gaussian function (compare Fig. 4.11 (j)). The weighting function is chosen so that the sine bell fits exactly across the acquisition time; mathematically the required function is:

$$W(t) = \sin\left(\frac{\pi t}{t_{\text{acq}}}\right).$$

The sine bell can be modified by shifting it the left, as is shown in Fig. 4.13. The further the shift to the left the smaller the resolution enhancement effect will be, and in the limit that the shift is by $\pi/2$ or 90° the function is simply a decaying one and so will broaden the lines. The shift is usually expressed in terms of a phase ϕ (in radians); the resulting weighting function is:

$$W(t) = \sin\left(\frac{(\pi - \phi)t}{t_{\text{acq}}} + \phi\right).$$

Note that this definition of the function ensures that it goes to zero at t_{acq} .

The shape of all of these weighting functions are altered subtly by squaring them to give the *sine bell squared* functions; these are also shown in Fig. 4.13. The weighting function is then

$$W(t) = \sin^2\left(\frac{(\pi - \phi)t}{t_{\text{acq}}} + \phi\right).$$

Much of the popularity of these functions probably rests of the fact that there is only one parameter to adjust, rather than two in the case of the Lorentz-to-Gauss transformation.

4.7 Zero filling

Before being processed the FID must be converted into a digital form so that it can be stored in computer memory. We will have more to say about this in Chapter 5 but for now we will just note that in this process the signal is sampled at regular intervals. The FID is therefore represented by a series of *data points*. When the FID is Fourier transformed the spectrum is also represented by a series of data points. So, although we plot the spectrum as a smooth line, it is in fact a series of closely spaced points.

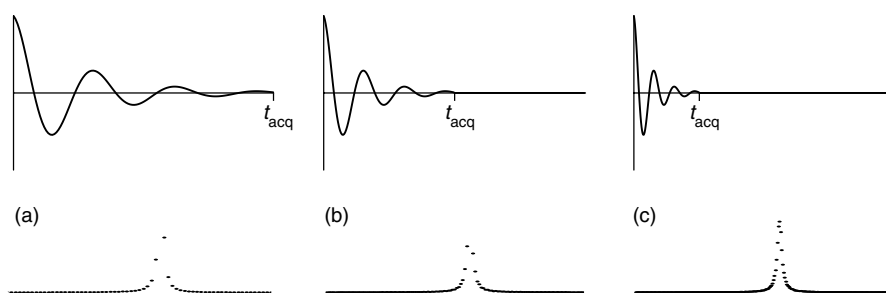


Fig. 4.14 Illustration of the results of zero filling. The FIDs along the top row have been supplemented with increasing numbers of zeroes and so contain more and more data points. Fourier transformation preserves the number of data points so the line in the spectrum is represented by more points as zeroes are added to the end of the FID. Note that the FID remains the same for all three cases; no extra data has been acquired.

This is illustrated in Fig. 4.14 (a) which shows the FID and the corresponding spectrum; rather than joining up the points which make up the spectrum we have just plotted the points. We can see that there are only a few data points which define the line in the spectrum.

If we take the original FID and add an equal number of zeroes to it, the corresponding spectrum has double the number of points and so the line is represented by more data points. This is illustrated in Fig. 4.14 (b). Adding a set of zeroes equal to the number of data points is called “one zero filling”.

We can carry on with this zero filling process. For example, having added one set of zeroes, we can add another to double the total number of data points (“two zero fillings”). This results in an even larger number of data points defining the line, as is shown in Fig. 4.14 (c).

Zero filling costs nothing in the sense that no extra data is required; it is just a manipulation in the computer. Of course, it does not improve the resolution as the measured signal remains the same, but the lines will be better defined in the spectrum. This is desirable, at least for aesthetic reasons if nothing else!

It turns out that the Fourier transform algorithm used by computer programs is most suited to a number of data points which is a power of 2. So, for example, $2^{14} = 16384$ is a suitable number of data points to transform, but 15000 is not. In practice, therefore, it is usual to zero fill the time domain data so that the total number of points is a power of 2; it is always an option, of course, to zero fill beyond this point.

4.8 Truncation

In conventional NMR it is virtually always possible to record the FID until it has decayed almost to zero (or into the noise). However, in multi-dimensional NMR this may not be the case, simply because of the restrictions on the amount of data which can be recorded, particularly in the “indirect dimension” (see Chapter X for further details). If we stop recording the signal before it has fully decayed the FID is said to be “truncated”; this is illustrated in Fig. 4.15.

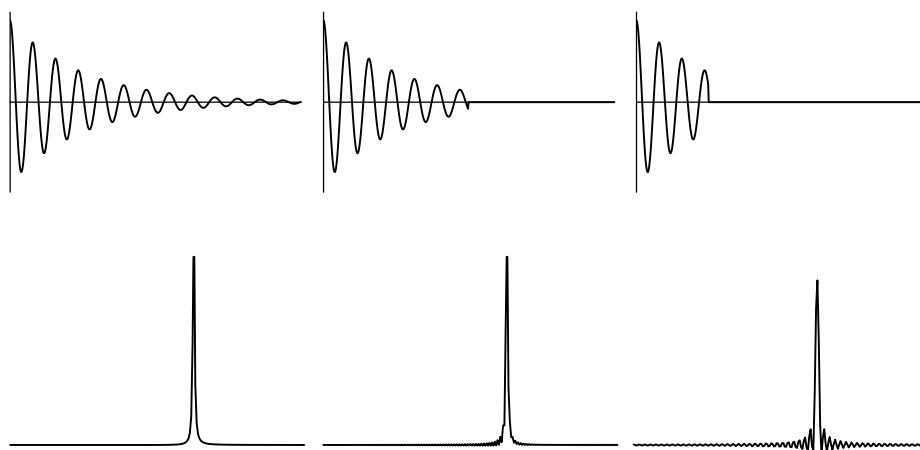


Fig. 4.15 Illustration of how truncation leads to artefacts (called *sinc wiggles*) in the spectrum. The FID on the left has been recorded for sufficient time that it has decayed almost to zero; the corresponding spectrum shows the expected lineshape. However, if data recording is stopped before the signal has fully decayed the corresponding spectra show oscillations around the base of the peak.

As is shown clearly in the figure, a truncated FID leads to oscillations around the base of the peak; these are usually called *sinc wiggles* or *truncation artefacts* – the name arises as the peak shape is related to a sinc function. The more severe the truncation, the larger the sinc wiggles. It is easy to show that the separation of successive maxima in these wiggles is $1/t_{\text{acq}}$ Hz.

Clearly these oscillations are undesirable as they may obscure nearby weaker peaks. Assuming that it is not an option to increase the acquisition time, the way forward is to apply a decaying weighting function to the FID so as to force the signal to go to zero at the end. Unfortunately, this will have the side effects of broadening the lines and reducing the SNR.

Highly truncated time domain signals are a feature of three- and higher-dimensional NMR experiments. Much effort has therefore been put into finding alternatives to the Fourier transform which will generate spectra without these truncation artefacts. The popular methods are *maximum entropy*, *linear prediction* and *FDM*. Each has its merits and drawbacks; they all need to be applied with great care.

4.9 Exercises

E 4-1

In a spectrum with just one line, the dispersion mode lineshape might be acceptable – in fact we can think of reasons why it might even be desirable (what might these be?). However, in a spectrum with many lines the dispersion mode lineshape is very undesirable – why?

E 4-2

Suppose that we record a spectrum with the simple pulse-acquire sequence using a 90° pulse applied along the x axis. The resulting FID is Fourier transformed and the spectrum is phased to give an absorption mode lineshape.

We then change the phase of the pulse from x to y , acquire an FID in the same way and phase the spectrum using the *same* phase correction as above. What lineshape would you expect to see in the spectrum; give the reasons for your answer.

How would the spectrum be affected by: (a) applying the pulse about $-x$; (b) changing the pulse flip angle to 270° about x ?

E 4-3

The gyromagnetic ratio of phosphorus-31 is $1.08 \times 10^8 \text{ rad s}^{-1} \text{ T}^{-1}$. This nucleus shows a wide range of shifts, covering some 700 ppm.

Suppose that the transmitter is placed in the middle of the shift range and that a 90° pulse of width $20 \mu\text{s}$ is used to excite the spectrum. Estimate the size of the phase correction which will be needed at the edges of the spectrum. (Assume that the spectrometer has a B_0 field strength of 9.4 T).

E 4-4

Why is it undesirable to continue to acquire the FID after the signal has decayed away?

How can weighting functions be used to improve the SNR of a spectrum? In your answer described how the parameters of a suitable weighting function can be chosen to optimize the SNR. Are there any disadvantages to the use of such weighting functions?

E 4-5

Describe how weighting functions can be used to improve the resolution in a spectrum. What sets the limit on the improvement that can be obtained in practice? Is zero filling likely to improve the situation?

E 4-6

Explain why use of a sine bell weighting function shifted by 45° may enhance the resolution but use of a sine bell shifted by 90° does not.

E 4-7

In a proton NMR spectrum the peak from TMS was found to show “wig-

gles” characteristic of truncation of the FID. However, the other peaks in the spectrum showed no such artefacts. Explain.

How can truncation artefacts be suppressed? Mention any difficulties with your solution to the problem.

5 How the spectrometer works

NMR spectrometers have now become very complex instruments capable of performing an almost limitless number of sophisticated experiments. However, the really important parts of the spectrometer are not that complex to understand in outline, and it is certainly helpful when using the spectrometer to have some understanding of how it works.

Broken down to its simplest form, the spectrometer consists of the following components:

- An intense, homogeneous and stable magnetic field
- A “probe” which enables the coils used to excite and detect the signal to be placed close to the sample
- A high-power RF transmitter capable of delivering short pulses
- A sensitive receiver to amplify the NMR signals
- A digitizer to convert the NMR signals into a form which can be stored in computer memory
- A “pulse programmer” to produce precisely times pulses and delays
- A computer to control everything and to process the data

We will consider each of these in turn.

5.1 The magnet

Modern NMR spectrometers use persistent superconducting magnets to generate the B_0 field. Basically such a magnet consists of a coil of wire through which a current passes, thereby generating a magnetic field. The wire is of a special construction such that at low temperatures (less than 6 K, typically) the resistance goes to zero – that is the wire is *superconducting*. Thus, once the current is set running in the coil it will persist for ever, thereby generating a magnetic field without the need for further electrical power. Superconducting magnets tend to be very stable and so are very useful for NMR.

To maintain the wire in its superconducting state the coil is immersed in a bath of liquid helium. Surrounding this is usually a “heat shield” kept at 77 K by contact with a bath of liquid nitrogen; this reduces the amount of (expensive) liquid helium which boils off due to heat flowing in from the surroundings. The whole assembly is constructed in a vacuum flask so as to further reduce the heat flow. The cost of maintaining the magnetic field is basically the cost of liquid helium (rather expensive) and liquid nitrogen (cheap).

Of course, we do not want the sample to be at liquid helium temperatures, so a room temperature region – accessible to the outside world – has to be engineered as part of the design of the magnet. Usually this room temperature zone takes the form of a vertical tube passing through the magnet (called the *bore tube* of the magnet); the magnetic field is in the direction of this tube.

Shims

The lines in NMR spectra are very narrow – linewidths of 1 Hz or less are not uncommon – so the magnetic field has to be very homogeneous, meaning that it must not vary very much over space. The reason for this is easily demonstrated by an example.

Consider a proton spectrum recorded at 500 MHz, which corresponds to a magnetic field of 11.75 T. Recall that the Larmor frequency is given by

$$\nu_0 = -\frac{1}{2\pi}\gamma B_0 \quad (5.1)$$

where γ is the gyromagnetic ratio ($2.67 \times 10^8 \text{ rad s}^{-1} \text{ T}^{-1}$ for protons). We need to limit the variation in the magnetic field across the sample so that the corresponding variation in the Larmor frequency is much less than the width of the line, say by a factor of 10.

With this condition, the maximum acceptable change in Larmor frequency is 0.1 Hz and so using Eq. 5.1 we can compute the change in the magnetic field as $2\pi 0.1/\gamma = 2.4 \times 10^{-9} \text{ T}$. Expressed as a fraction of the main magnetic field this is about 2×10^{-10} . We can see that we need to have an extremely homogeneous magnetic field for work at this resolution.

On its own, no superconducting magnet can produce such a homogeneous field. What we have to do is to surround the sample with a set of *shim coils*, each of which produces a tiny magnetic field with a particular spatial profile. The current through each of these coils is adjusted until the magnetic field has the required homogeneity, something we can easily assess by recording the spectrum of a sample which has a sharp line. Essentially how this works is that the magnetic fields produced by the shims are cancelling out the small residual inhomogeneities in the main magnetic field.

Modern spectrometers might have up to 40 different shim coils, so adjusting them is a very complex task. However, once set on installation it is usually only necessary on a day to day basis to alter a few of the shims which generate the simplest field profiles.

The shims are labelled according to the field profiles they generate. So, for example, there are usually shims labelled x , y and z , which generate magnetic fields varying the corresponding directions. The shim z^2 generates a field that varies quadratically along the z direction, which is the direction of B_0 . There are more shims whose labels you will recognize as corresponding to the names of the hydrogen atomic orbitals. This is no coincidence; the magnetic field profiles that the shim coils create are in fact the spherical harmonic functions, which are the angular parts of the atomic orbitals.

5.2 The probe

The probe is a cylindrical metal tube which is inserted into the bore of the magnet. The small coil used to both excite and detect the NMR signal (see sections 3.2 and 3.4) is held in the top of this assembly in such a way that the sample can come down from the top of the magnet and drop into the coil. Various other pieces of electronics are contained in the probe, along with some arrangements for heating or cooling the sample.

The key part of the probe is the small coil used to excite and detect the magnetization. To optimize the sensitivity this coil needs to be as close as possible to the sample, but of course the coil needs to be made in such a way that the sample tube can drop down from the top of the magnet into the coil. Extraordinary effort has been put into the optimization of the design of this coil.

The coil forms part of a *tuned circuit* consisting of the coil and a capacitor. The inductance of the coil and the capacitance of the capacitor are set such that the tuned circuit they form is resonant at the Larmor frequency. That the coil forms part of a tuned circuit is very important as it greatly increases the detectable current in the coil.

Spectroscopists talk about “tuning the probe” which means adjusting the capacitor until the tuned circuit is resonant at the Larmor frequency. Usually we also need to “match the probe” which involves further adjustments designed to maximize the power transfer between the probe and the transmitter and receiver; Fig. 5.1 shows a typical arrangement. The two adjustments tend to interact rather, so tuning the probe can be a tricky business. To aid us, the instrument manufacturers provide various indicators and displays so that the tuning and matching can be optimized. We expect the tuning of the probe to be particularly sensitive to changing solvent or to changing the concentration of ions in the solvent.

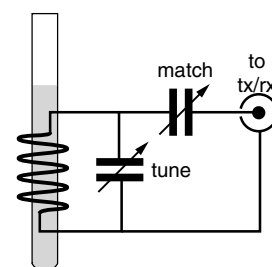


Fig. 5.1 Schematic of the key parts of the probe. The coil is shown on the left (with the sample tube in grey); a tuned circuit is formed by a capacitor (marked “tune”). The power transfer to the transmitter and receiver (tx. and rx.) is optimized by adjusting the capacitor marked “match”. Note that the coil geometry as shown is not suitable for a superconducting magnet in which the main field is parallel to the sample axis.

5.3 The transmitter

The radiofrequency transmitter is the part of the spectrometer which generates the pulses. We start with an RF source which produces a stable frequency which can be set precisely. The reason why we need to be able to set the frequency is that we might want to move the transmitter to different parts of the spectrum, for example if we are doing experiments involving selective excitation (section 3.11).

Usually a *frequency synthesizer* is used as the RF source. Such a device has all the desirable properties outlined above and is also readily controlled by a computer interface. It is also relatively easy to phase shift the output from such a synthesizer, which is something we will need to do in order to create phase shifted pulses.

As we only need the RF to be applied for a short time, the output of the synthesizer has to be “gated” so as to create a pulse of RF energy. Such a gate will be under computer control so that the length and timing of the pulse can be controlled.

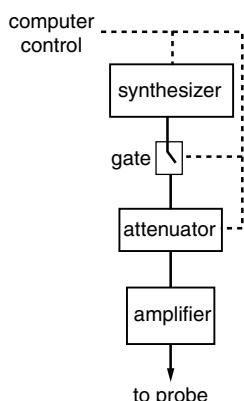


Fig. 5.2 Typical arrangement of the RF transmitter. The synthesizer which is the source of the RF, the attenuator and the gate used to create the pulses all under computer control.

The RF source is usually at a low level (a few mW) and so needs to be boosted considerably before it will provide a useful B_1 field when applied to the probe; the complete arrangement is illustrated in Fig. 5.2. RF amplifiers are readily available which will boost this small signal to a power of 100 W or more. Clearly, the more power that is applied to the probe the more intense the B_1 field will become and so the shorter the 90° pulse length. However, there is a limit to the amount of power which can be applied because of the high voltages which are generated in the probe.

When the RF power is applied to the tuned circuit of which the coil is part, high voltages are generated across the tuning capacitor. Eventually, the voltage will reach a point where it is sufficient to ionize the air, thus generating a discharge or arc (like a lightning bolt). Not only does this *probe arcing* have the potential to destroy the coil and capacitor, but it also results in unpredictable and erratic B_1 fields. Usually the manufacturer states the power level which is “safe” for a particular probe.

Power levels and “dB”

The spectrometer usually provides us with a way of altering the RF power level and hence the strength of the B_1 field. This is useful as we may wish to set the B_1 field strength to a particular level, for example for selective excitation.

The usual way of achieving this control is add an *attenuator* between the RF source (the synthesizer) and the amplifier. As its name implies, the attenuator reduces the signal as it passes through.

The attenuation is normally expressed in *decibels* (abbreviated dB and pronounced “dee-bee”). If the input power is P_{in} and the output power is P_{out} the attenuation in dB is

$$10 \times \log_{10} \frac{P_{out}}{P_{in}};$$

note that the logarithm is to the base 10, not the natural logarithm. The factor of 10 is the “deci” part in the dB.

For example, if the output power is half the input power, i.e. $P_{in} = 2 \times P_{out}$, the power ratio in dB is

$$10 \times \log_{10} \frac{P_{out}}{P_{in}} = 10 \times \log_{10} \frac{1}{2} = -3.0$$

So, halving the power corresponds to a change of -3.0 dB, the minus indicating that there is a power reduction i.e. an attenuation. An attenuator which achieves this effect would be called “a 3 dB attenuator”.

Likewise, a power reduction by a factor of 4 corresponds to -6.0 dB. In fact, because of the logarithmic relationship we can see that each 3 dB of attenuation will halve the power. So, a 12 dB attenuator will reduce the power by a factor of 16.

It turns out that the B_1 field strength is proportional to the square root of the power applied. The reason for this is that it is the current in the coil which is responsible for the B_1 field and, as we know from elementary electrical theory, power is equal to (resistance $\times$ current²). So, the current is proportional to the square root of the power.

Therefore, in order to double the B_1 field we need to double the current and this means multiplying the power by a factor of four. Recall from the above that a change in the power by a factor of 4 corresponds to 6 dB. Thus, 6 dB of attenuation causes the B_1 field to fall to half its original value; 12 dB would cause it to fall to one quarter of its initial value, and so on.

Usually the attenuator is under computer control and its value can be set in dB. This is very helpful to us as we can determine the attenuation needed for different B_1 field strengths. Suppose that with a certain setting of the attenuator we have determined the B_1 field strength to be $\omega_1^{\text{init}}/2\pi$, where we have expressed the field strength in frequency units as described in section 3.4. However, in our experiment we want the field strength to be $\omega_1^{\text{new}}/2\pi$. The ratio of the powers needed to achieve these two field strengths is equal to the square of the ratio of the field strengths:

$$\text{power ratio} = \left(\frac{\omega_1^{\text{new}}/2\pi}{\omega_1^{\text{init}}/2\pi} \right)^2.$$

Expressed in dB this is

$$\begin{aligned} \text{power ratio in dB} &= 10 \log_{10} \left(\frac{\omega_1^{\text{new}}/2\pi}{\omega_1^{\text{init}}/2\pi} \right)^2 \\ &= 20 \log_{10} \left(\frac{\omega_1^{\text{new}}/2\pi}{\omega_1^{\text{init}}/2\pi} \right). \end{aligned}$$

This expression can be used to find the correct setting for the attenuator.

As the duration of a pulse of a given flip angle is *inversely* proportional to $\omega_1/2\pi$ the relationship can be expressed in terms of the initial and new pulse widths, τ_{init} and τ_{new} :

$$\text{power ratio in dB} = 20 \log_{10} \left(\frac{\tau_{\text{init}}}{\tau_{\text{new}}} \right).$$

For example, suppose we have calibrated the pulse width for a 90° pulse to be $15 \mu\text{s}$ but what we actually wanted was a 90° pulse of $25 \mu\text{s}$. The required attenuation would be:

$$\text{power ratio in dB} = 20 \log_{10} \left(\frac{15}{25} \right) = -4.4 \text{ dB}.$$

We would therefore need to increase the attenuator setting by 4.4 dB.

Phase shifted pulses

In section 3.9 we saw that pulses could be applied along different axes in the rotating frame, for example x and y . At first sight you might think that to achieve this we would need a coil oriented along the y axis to apply a pulse along y . However, this is not the case; all we need to do is to shift the phase of the RF by 90° .

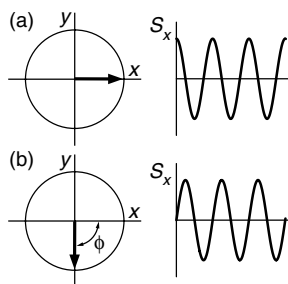


Fig. 5.3 Illustration of how a 90° phase shift takes us from cosine to sine modulation. In (a) the vector starts out along x ; as it rotates in a positive sense the x component, S_x , varies as a cosine wave, as is shown in the graph on the right. In (b) the vector starts on $-y$; the x component now takes the form of a sine wave, as is shown in the graph. Situation (b) is described as a phase shift, ϕ , of -90° compared to (a).

In section 3.4 we described the situation where RF power at frequency ω_{RF} is applied to the coil. In this situation the B_1 field along the x axis can be written as

$$2B_1 \cos \omega_{\text{RF}} t.$$

If we phase shift the RF by 90° the field will be modulated by a sine function rather than a cosine:

$$2B_1 \sin \omega_{\text{RF}} t.$$

How this comes about is illustrated in Fig. 5.3.

The difference between the cosine and sine modulations is that in the former the field starts at a maximum and then decays, whereas in the latter it starts at zero and increases.

The resulting sine modulated field can still be thought of as arising from two counter-rotating fields, B_1^+ and B_1^- ; this is shown in Fig. 5.4. The crucial thing is that as the field is zero at time zero the two counter-rotating vectors start out opposed. As they evolve from this position the component along x grows, as required. Recall from section 3.4 that it is the component B_1^- which interacts with the magnetization. When we move to a rotating frame at ω_{RF} this component will be along the y axis (the position it has at time zero). This is in contrast to the case in Fig. 3.8 where B_1^- starts out on the x axis.

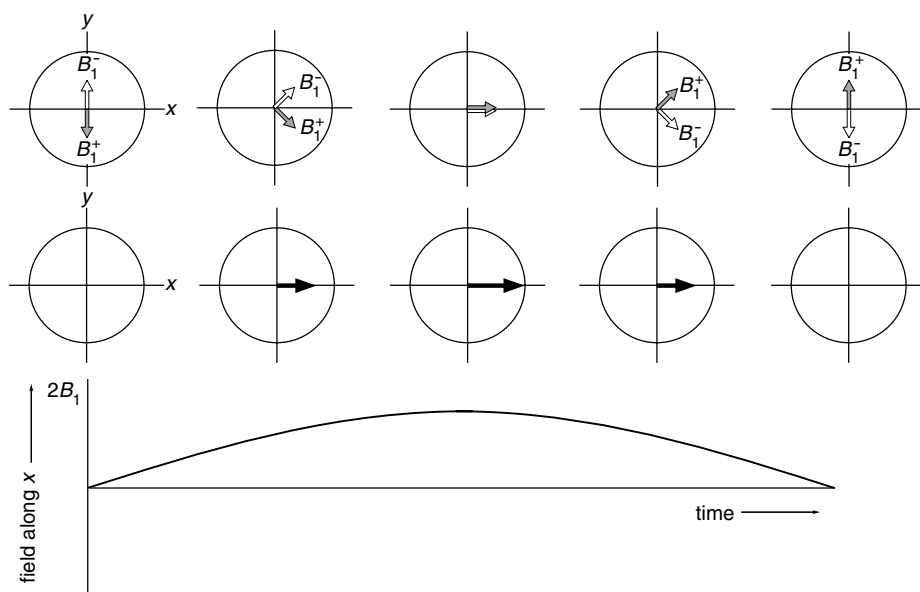


Fig. 5.4 Illustration of the decomposition of a field oscillating along the x axis according to a sine wave into two counter rotating components. This figure should be compared to Fig. 3.8. Note that the component B_1^- starts out along the y axis at time zero in contrast to the case of Fig. 3.8 where it starts out along x .

So, if the B_1 field is modulated by a cosine wave the result is a pulse about x , whereas if it is modulated by a sine wave the result is a pulse about y . We see that by phase shifting the RF we can apply pulses about any axis; there is no need to install more than one coil in the probe.

5.4 The receiver

The NMR signal emanating from the probe is very small (of the order of μV) but for modern electronics there is no problem in amplifying this signal to a level where it can be digitized. These amplifiers need to be designed so that they introduce a minimum of extra noise (they should be *low-noise* amplifiers).

The first of these amplifiers, called the *pre-amplifier* or *pre-amp* is usually placed as close to the probe as possible (you will often see it resting by the foot of the magnet). This is so that the weak signal is boosted before being sent down a cable to the spectrometer console.

One additional problem which needs to be solved comes about because the coil in the probe is used for both exciting the spins and detecting the signal. This means that at one moment hundreds of Watts of RF power are being applied and the next we are trying to detect a signal of a few μV . We need to ensure that the high-power pulse does not end up in the sensitive receiver, thereby destroying it!

This separation of the receiver and transmitter is achieved by a gadget known as a *diplexer*. There are various different ways of constructing such a device, but at the simplest level it is just a fast acting switch. When the pulse is on the high power RF is routed to the probe and the receiver is protected by disconnecting it or shorting it to ground. When the pulse is off the receiver is connected to the probe and the transmitter is disconnected.

Some diplexers are passive in the sense that they require no external power to achieve the required switching. Other designs use fast electronic switches (rather like the gate in the transmitter) and these are under the command of the pulse programmer so that the receiver or transmitter are connected to the probe at the right times.

5.5 Digitizing the signal

The analogue to digital converter

A device known as an *analogue to digital converter* or ADC is used to convert the NMR signal from a voltage to a binary number which can be stored in computer memory. The ADC samples the signal at regular intervals, resulting in a representation of the FID as *data points*.

The output from the ADC is just a number, and the largest number that the ADC can output is set by the number of binary “bits” that the ADC uses. For example with only three bits the output of the ADC could take just 8 values: the binary numbers 000, 001, 010, 011, 100, 101, 110 and 111. The smallest number is 0 and the largest number is decimal 7 (the total number of possibilities is 8, which is 2 raised to the power of the number of bits). Such an ADC would be described as a “3 bit ADC”.

The waveform which the ADC is digitizing is varying continuously, but output of the 3 bit ADC only has 8 levels so what it has to do is simply output the level which is closest to the current input level; this is illustrated in Fig. 5.5. The numbers that the ADC outputs are therefore an approximation

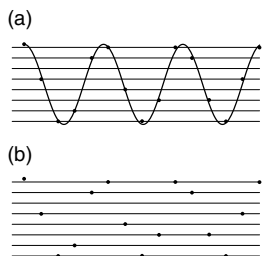


Fig. 5.5 Digitization of a waveform using an ADC with 8 levels (3 bits). The output of the ADC can only be one of the 8 levels, so the smoothly varying waveform has to be approximated by data points at one of the 8 levels. The data points, indicated by filled circles, are therefore an approximation to the true waveform.

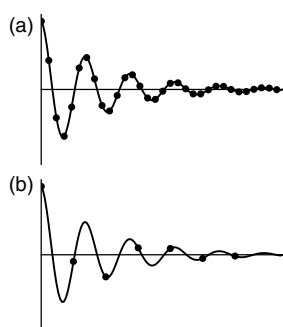


Fig. 5.7 Illustration of the effect of sampling rate on the representation of the FID. In (a) the data points (shown by dots) are quite a good representation of the signal (shown by the continuous line). In (b) the data points are too widely separated and so are a very poor representation of the signal.

to the actual waveform.

The approximation can be improved by increasing the number of bits; this gives more output levels. At present, ADC with between 16 and 32 bits are commonly in use in NMR spectrometers. The move to higher numbers of bits is limited by technical considerations.

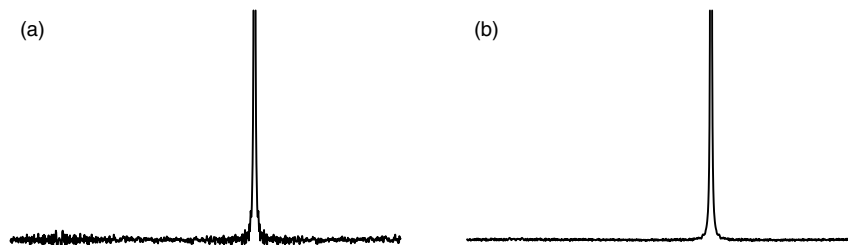


Fig. 5.6 Spectrum (a) is from an FID which has been digitized using a 6 bit ADC (i.e. 64 levels); the vertical scale has been expanded 10 fold so that the digitization sidebands are clearly visible. Spectrum (b) is from an FID which has been digitized using an 8 bit ADC (256 levels); the improvement over (a) is evident.

The main consequence of the approximation process which the ADC uses is the generation of a forest of small sidebands – called *digitization sidebands* – around the base of the peaks in the spectrum. Usually these are not a problem as they are likely to be swamped by noise. However, if the spectrum contains a very strong peak the sidebands from it can swamp a nearby weak peak. Improving the resolution of the ADC, i.e. the number of bits it uses, reduces the digitization sidebands, as is shown in Fig. 5.6.

Sampling rates

Given that the ADC is only going to sample the signal at regular intervals the question arises as to how frequently it is necessary to sample the FID i.e. what should the time interval between the data points be. Clearly, if the time interval is too long we will miss crucial features of the waveform and so the digitized points will be a poor representation of the signal. This is illustrated in Fig. 5.7.

It turns out that if the interval between the points is Δ the highest frequency which can be represented correctly, $f_{\max}$, is given by

$$f_{\max} = \frac{1}{2\Delta};$$

$f_{\max}$ is called the *Nyquist frequency*. Usually we think of this relationship the other way round i.e. if we wish to represent correctly frequencies up to $f_{\max}$ the sampling interval is given by:

$$\Delta = \frac{1}{2f_{\max}}.$$

This sampling interval is often called the *dwell time*. A signal at $f_{\max}$ will have two data points per cycle.

We will see shortly that we are able to distinguish positive and negative frequencies, so a dwell time of Δ means that the range of frequencies from $-f_{\max}$ to $+f_{\max}$ are represented correctly.

A signal at greater than $f_{\max}$ will still appear in the spectrum but not at the correct frequency; such a peak is said to be *folded*. For example a peak at $(f_{\max} + F)$ will appear in the spectrum at $(-f_{\max} + F)$ as is illustrated in Fig. 5.8.

This Nyquist condition quickly brings us to a problem. A typical NMR frequency is of the order of hundreds of MHz but there simply are no ADCs available which work fast enough to digitize such a waveform with the kind of accuracy (i.e. number of bits) we need for NMR. The solution to this problem is to mix down the signal to a lower frequency, as is described in the next section.

Mixing down to a lower frequency

Luckily for us, although the NMR frequency is quite high, the range of frequencies that any typical spectrum covers is rather small. For example, the 10 ppm of a proton spectrum recorded at 400 MHz covers just 4000 Hz. Such a small range of frequencies is easily within the capability of suitable ADCs.

The procedure is to take the NMR signals and then *subtract* a frequency from them all such that we are left with a much lower frequency. Typically, the frequency we subtract is set somewhere in the middle of the spectrum; this frequency is called the *receiver reference frequency* or just the *receiver frequency*.

For example, suppose that the 10 ppm range of proton shifts runs from 400.000 MHz to 400.004 MHz. If we set the receiver reference frequency at 400.002 MHz and then subtract this from the NMR frequencies we end up with a range -2000 to $+2000$ Hz.

The subtraction process is carried out by a device called a *mixer*. There are various ways of actually making a mixer – some are “active” and contain the usual transistors and so on, some are passive and are mainly constructed from transformers and diodes. The details need not concern us – all we need to know is that a mixer takes two signal inputs at frequencies f_1 and f_2 and produces an output signal which contains the signals at the sum of the two inputs, $(f_1 + f_2)$ and the difference $(f_1 - f_2)$. The process is visualized in Fig. 5.9.

Usually one of the inputs is generated locally, typically by a synthesizer. This signal is often called the *local oscillator*. The other input is the NMR signal the probe. In the spectrometer the local oscillator is set to a required receiver frequency so that the difference frequency will be quite small. The other output of the mixer is the sum frequency, which will be at the order of twice the Larmor frequency. This high frequency signal is easily separated from the required low frequency signal by a passing the output of the mixer through a *low pass filter*. The filtered signal is then passed to the ADC.

5.6 Quadrature detection

In the previous section we saw that the signal which is passed to the ADC contains positive and negative frequencies – essentially this corresponds to positive and negative offsets. The question therefore arises as to whether or

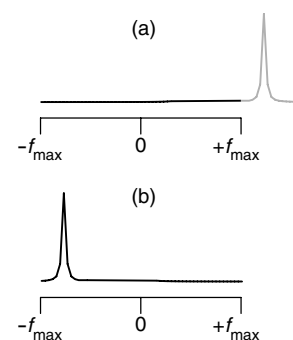


Fig. 5.8 Illustration of the concept of folding. In spectrum (a) the peak (shown in grey) is at a higher frequency than the maximum set by the Nyquist condition. In practice, such a peak would appear in the position shown in (b).

not we are able to discriminate between frequencies which only differ in their sign; the answer is yes, but only provided we use a method known as *quadrature detection*.

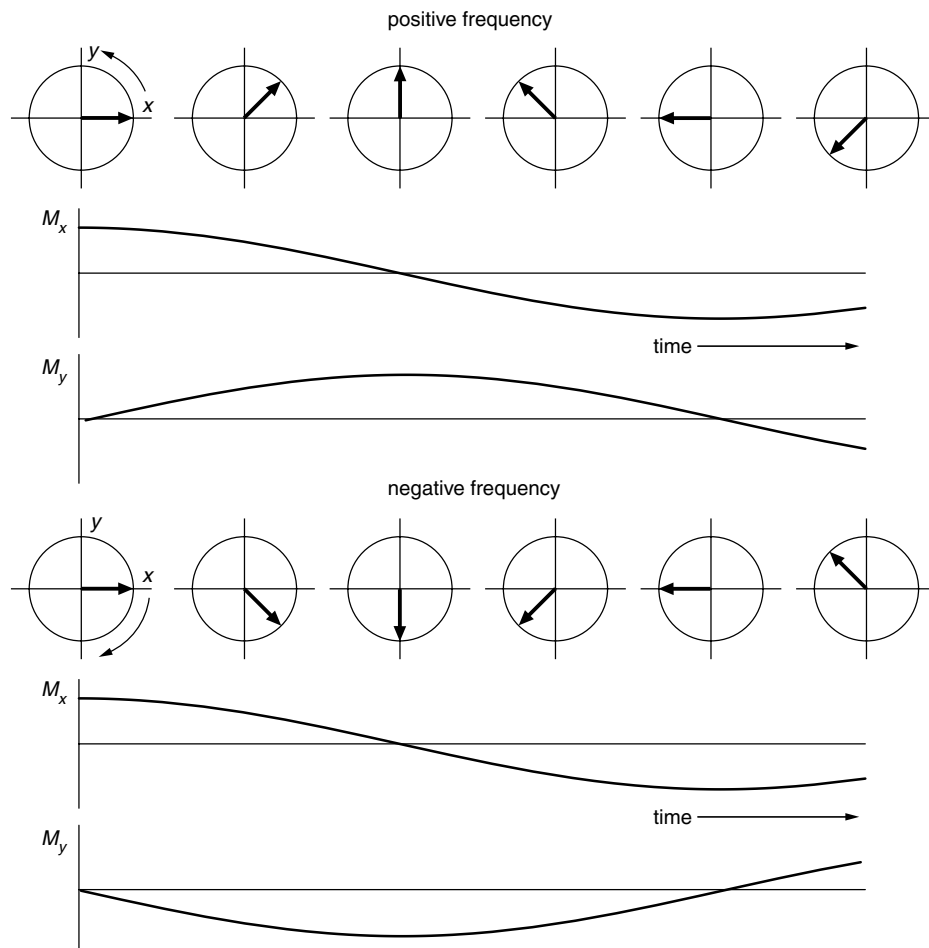


Fig. 5.10 Illustration of the x and y components from a vector precessing as a positive and negative offset. The only difference between the two is in the y component.

Figure 5.10 illustrates the problem. The upper part shows a vector precessing with a positive offset frequency (in the rotating frame); the x component is a cosine wave and the y component is a sine wave. The lower part shows the evolution of a vector with a negative frequency. The x component remains the same but the y component has changed sign when compared with evolution at a positive frequency. We thus see that to distinguish between positive and negative frequencies we need to know *both* the x and y components of the magnetization.

It is tempting to think that it would be sufficient just to know the y component as positive frequencies give a positive sine wave and negative ones give a negative sine wave. In the corresponding spectrum this would mean that peaks at positive offsets would be positive and those at negative offsets would be negative. If there were a few well separated peaks in the spectrum such a spectrum might be interpretable, but if there are many peaks in the spectrum

the result would be an uninterpretable mess.

How then can we detect both the x and y components of the magnetization? One idea is that we should have two coils, one along x and one along y : these would certainly detect the x and y components. In practice, it turns out to be very hard to achieve such an arrangement, partly because of the confined space in the probe and partly because of the difficulties in making the two coils electrically isolated from one another.

Luckily, though, it turns out to be easy to achieve the same effect by phase shifting the receiver reference frequency; we describe how this works in the next section.

The mixing process

In section 5.5 we described the use of a mixer to subtract a locally generated frequency from that of the NMR signal. The mixer achieves this by *multiplying* together the two input signals, and it is instructive to look at the consequence of this in more detail.

Suppose that the signal coming from the probe can be written as $A \cos \omega_0 t$ where A gives the overall intensity and as usual ω_0 is the Larmor frequency. Let us write the local oscillator signal as $\cos \omega_{\text{rx}} t$, where ω_{rx} is the receiver frequency. Multiplying these two together gives:

$$A \cos \omega_0 t \times \cos \omega_{\text{rx}} t = \frac{1}{2} A [\cos(\omega_0 + \omega_{\text{rx}})t + \cos(\omega_0 - \omega_{\text{rx}})t]$$

where we have used the well known formula $\cos A \cos B = \frac{1}{2}(\cos(A + B) + \cos(A - B))$.

After the low pass filter only the term $\cos(\omega_0 - \omega_{\text{rx}})t$ will survive. We recognize the difference frequency $(\omega_0 - \omega_{\text{rx}})$ as the separation between the receiver frequency and the Larmor frequency. If the receiver frequency is made to be the same as the transmitter frequency (as is usually the case) this difference frequency is the offset we identified in section 3.4.

In summary, the output of the mixer is at the offset frequency in the rotating frame and takes the form of a cosine modulation; it is thus equivalent to detecting the x component of the magnetization in the rotating frame.

Now consider the case where the local oscillator signal is $-\sin \omega_{\text{rx}} t$; as we saw in section 5.3 such a signal can be created by a phase shift of 90° . The output of the mixer is now given by

$$A \cos \omega_0 t \times -\sin \omega_{\text{rx}} t = \frac{1}{2} A [-\sin(\omega_0 + \omega_{\text{rx}})t + \sin(\omega_0 - \omega_{\text{rx}})t]$$

where we have used the well known formula $\cos A \sin B = \frac{1}{2}(\sin(A + B) - \sin(A - B))$. We now see that the difference frequency term is sine modulated at the offset frequency; it is the equivalent of the y component in the rotating frame. Thus, simply by shifting the phase of the local oscillator signal we can detect both the x and y components in the rotating frame.

The complete arrangement for quadrature detection is outlined in Fig. 5.11. The NMR signal from the probe is split into two and fed to two separate mixers. The local oscillator signals fed to the two mixers are phase shifted by 90° relative to one another; as a result, the outputs of the two mixers

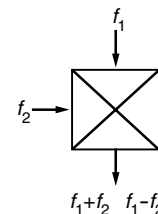


Fig. 5.9 A radiofrequency mixer takes inputs at two different frequencies and produces an output which contains the sum and difference of the two inputs.

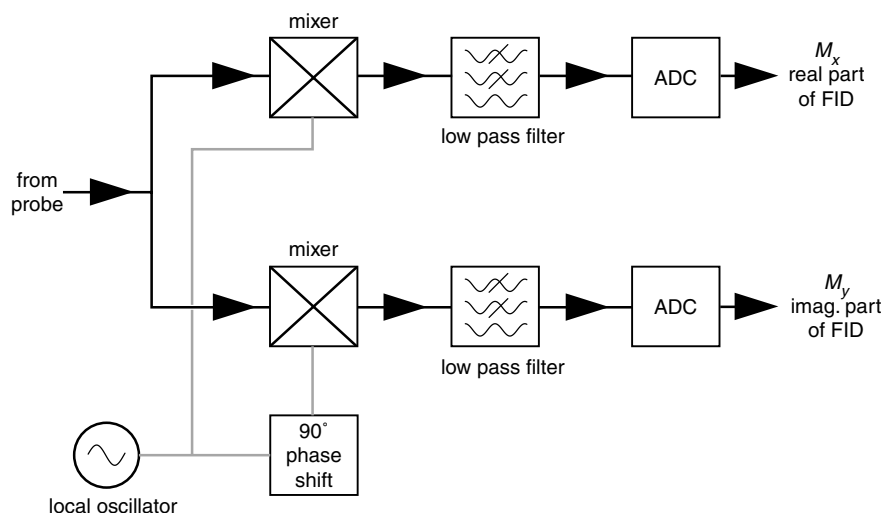


Fig. 5.11 Schematic of the arrangement for quadrature detection; see text for details.

are the equivalents of the x and y components in the rotating frame. These two outputs are digitized separately and become the real and imaginary parts of a complex time domain signal (as described in section 4.1). Such a quadrature detection scheme is the norm for all modern NMR spectrometers.

5.7 Time and frequency

We are now in a position to tidy up a number of points of terminology relating to the FID and spectrum. It is important to understand these in order to report correctly the parameters used to record a spectrum.

The range of frequencies correctly represented in the spectrum (i.e. not folded) is called the *spectral width*, f_{SW} . As quadrature detection is used, the frequency scale on the spectrum runs from $-\frac{1}{2}f_{\text{SW}}$ to $+\frac{1}{2}f_{\text{SW}}$ i.e. positive and negative frequencies are discriminated.

For a given spectral width, the sampling interval or dwell time, Δ , is computed from:

$$\Delta = \frac{1}{f_{\text{SW}}}$$

which means that the Nyquist frequency is $\frac{1}{2}f_{\text{SW}}$ – the edges of the spectrum.

If we record the FID for a time t_{acq} , called the acquisition time, then the number of data points, N , is

$$N = \frac{t_{\text{acq}}}{\Delta}.$$

Each data point is complex, consisting of a real and an imaginary part.

To be clear about the conditions under which an FID is recorded it is important to quote the acquisition time and the spectral width; quoting the number of data points on its own is meaningless.

5.8 The pulse programmer

The pulse programmer has become an immensely sophisticated piece of computer hardware, controlling as it does all of the functions of the spectrometer. As the pulse programmer needs to produce very precisely timed events, often in rapid succession, it is usual for it to run independently of the main computer. Typically, the pulse program is specified in the main computer and then, when the experiment is started, the instructions are loaded into the pulse programmer and then executed there.

The acquisition of data is usually handled by the pulse programmer, again separately from the main computer.

5.9 Exercises

E 5-1

You have been offered a superconducting magnet which claims to have a homogeneity of “1 part in 10^8 ”. Your intention is to use it to record phosphorus-31 spectra at Larmor frequency of 180 MHz, and you know that your typical linewidths are likely to be of the order of 25 Hz. Is the magnet sufficiently homogeneous to be of use?

E 5-2

A careful pulse calibration experiment determines that the 180° pulse is $24.8 \mu\text{s}$. How much attenuation, in dB, would have to be introduced into the transmitter in order to give a field strength, $(\omega_1/2\pi)$, of 2 kHz?

E 5-3

A spectrometer is equipped with a transmitter capable of generating a maximum of 100 W of RF power at the frequency of carbon-13. Using this transmitter at full power the 90° pulse width is found to be $20 \mu\text{s}$. What power would be needed to reduce the 90° pulse width to $7.5 \mu\text{s}$? Would you have any reservations about using this amount of power?

E 5-4

Explain what is meant by “a two bit ADC” and draw a diagram to illustrate the outcome of such a ADC being used to digitize a sine wave.

Why is it generally desirable to improve the number of bits that the ADC uses?

E 5-5

A spectrometer operates at 800 MHz for proton and it is desired to cover a shift range of 15 ppm. Assuming that the receiver frequency is placed in the middle of this range, what spectral width would be needed and what would the sampling interval (dwell time) have to be?

If we recorded a FID for 2 s with the spectral width set as you have determined, how many data points will have been collected?

E 5-6

Suppose that the spectral width is set to 100 Hz, but that a peak is present whose offset from the receiver is +60 Hz. Where will the peak appear in the spectrum? Can you explain why?